Motivic Homotopy Theory (Draft)

Ou Liu

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Preface

These notes grew out of my participation in Marc Hoyois's Oberseminar on motivic cohomology at the University of Regensburg (SoSe 2026), combined with my own study of the subject. The primary references are the work of Bachmann–Hoyois [BH21] on norms, and the recent paper of Bachmann–Elmanto–Morrow [BEM25].

The goal is to give a self-contained introduction to motivic homotopy theory.

Warning. *This document is a preliminary draft. It likely contains errors, incomplete arguments, and missing references. Corrections and suggestions are welcome.*

A Chinese version of part of this material is available on Banana Space.

Conventions

Throughout these notes, we work in the framework of ∞-categories. The word "category" means ∞-category unless otherwise specified; we write "ordinary category" or "1-category" for the classical notion.

All schemes are assumed to be quasi-compact and quasi-separated unless otherwise stated.

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Chapter 1.

Unstable Motivic Homotopy Theory

1.1. Introduction and Motivation

The original motivation for unstable motivic homotopy theory traces back to a question posed by Serre in [Ser55]: are all finitely generated projective modules over $k[x_1, \ldots, x_n]$ free? In the language of vector bundles, this asks whether the pullback map

$$\pi_0 \mathsf{Vect}_r(\mathrm{Spec}(k)) \to \pi_0 \mathsf{Vect}_r(\mathbb{A}^n_k)$$

is a bijection for any field k and integers $n, r \geq 1$. Quillen [Qui76] and Suslin [Sus76] independently answered this in the affirmative, proving the stronger statement that k can be replaced by any Dedekind domain. This naturally leads to the following:

Conjecture 1.1.1 (Bass–Quillen). *Let X be a regular Noetherian affine scheme of finite Krull dimension. Then for all $r, n \geq 1$, the pullback map*

$$\pi_0 \mathsf{Vect}_r(X) \to \pi_0 \mathsf{Vect}_r(X \times \mathbb{A}^n)$$

is a bijection.

This conjecture has been established in many cases—Lindel [**?**] proved it for schemes of finite type over a field, and Popescu handled the case of unramified regular local rings—but the full generality remains open.

The form of Conjecture 1.1.1 parallels a fundamental fact in topology: for any paracompact Hausdorff space X, there is a bijection

$$\pi_0 \mathsf{Vect}_r^{\mathrm{top}}(X) \xrightarrow{\sim} \pi_0 \mathsf{Vect}_r^{\mathrm{top}}(X \times I),$$

where $I = [0, 1]$ is the unit interval. Under this analogy, $\mathbb{A}^1$ plays the role of the interval I in algebraic geometry: both serve as "cylinder objects" in their respective categories. This suggests the possibility of a homotopy theory for algebraic geometry in which the projection $\mathbb{A}^1 \times X \to X$ is formally declared to be a homotopy equivalence.

Supporting this intuition is the empirical observation that many important cohomology theories on smooth schemes are $\mathbb{A}^1$-invariant. For a smooth scheme X over a field k:

- **Chow groups:** The pullback $\mathrm{CH}^*(X) \xrightarrow{\sim} \mathrm{CH}^*(X \times_k \mathbb{A}^1)$ is an isomorphism.
- **Grothendieck group:** The pullback $\mathrm{K}_0(X) \xrightarrow{\sim} \mathrm{K}_0(X \times_k \mathbb{A}^1)$ is an isomorphism.
- **Étale cohomology:** When ℓ is invertible in k, the pullback $\mathrm{H}^*_{\text{ét}}(X, \mu_\ell) \xrightarrow{\sim} \mathrm{H}^*_{\text{ét}}(X \times_k \mathbb{A}^1, \mu_\ell)$ is an isomorphism.

***Remark* 1.1.2** ($\mathbb{A}^1$-invariance of vector bundles: affine vs. non-affine). The functor $\pi_0 \mathsf{Vect}_r : \mathsf{Sm}_k^{\mathrm{op}} \to \mathsf{Set}$ is *not* $\mathbb{A}^1$-invariant on all smooth schemes. We exhibit a rank 2 bundle on $\mathbb{P}^1 \times \mathbb{A}^1$ which demonstrates the failure. Take the trivial rank 2 bundle on each of the standard charts $U_0 = \mathrm{Spec}(k[x]) \times \mathbb{A}^1$ and $U_1 = \mathrm{Spec}(k[x^{-1}]) \times \mathbb{A}^1$, and glue along

$$g_{01}(x, t) = \begin{pmatrix} x & t \\ 0 & x^{-1} \end{pmatrix} \in \mathrm{GL}_2(k[x, x^{-1}, t]).$$

Call the resulting bundle E. At $t = 0$, the transition matrix is diagonal, and $E|_{\mathbb{P}^1 \times \{0\}} \simeq \mathcal{O}(1) \oplus \mathcal{O}(-1)$. At $t = 1$, the transition matrix defines an extension

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow E|_{\mathbb{P}^1 \times \{1\}} \longrightarrow \mathcal{O}(1) \longrightarrow 0$$

whose class in $\mathrm{Ext}^1(\mathcal{O}(1), \mathcal{O}(-1)) = H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k$ is non-trivial; by Grothendieck's splitting theorem this extension, as a vector bundle, is isomorphic to $\mathcal{O} \oplus \mathcal{O}$. Thus $E|_{\mathbb{P}^1 \times \{0\}} \not\simeq E|_{\mathbb{P}^1 \times \{1\}}$ as vector bundles on $\mathbb{P}^1$. Since any bundle pulled back from $\mathbb{P}^1$ along the projection $\mathbb{P}^1 \times \mathbb{A}^1 \to \mathbb{P}^1$ has constant fibres up to isomorphism, E is not in the image of $\pi_0 \mathsf{Vect}_2(\mathbb{P}^1) \to \pi_0 \mathsf{Vect}_2(\mathbb{P}^1 \times \mathbb{A}^1)$.

Nevertheless, if we *force* Vect_r to be $\mathbb{A}^1$-invariant by applying motivic localization, the resulting functor still gives the correct values on *smooth affine schemes*. This is a deep result of Morel [Mor12], later simplified by Asok–Hoyois–Wendt [AHW17], and forms a cornerstone of the application of motivic homotopy theory to the classification of vector bundles.

1.1.1. Overview of the construction

Let S be a quasi-compact quasi-separated scheme, and let Sm_S denote the category of smooth S-schemes of finite presentation. Our goal is to construct the **motivic homotopy category** $\mathsf{H}(S)$: an enlargement of Sm_S in which the projections $\mathbb{A}^1 \times_S X \to X$ become isomorphisms.

The construction proceeds in three steps, each with clear geometric meaning:

(i) **Sifted cocompletion.** The category Sm_S lacks sufficient colimits for homotopy-theoretic constructions. We embed Sm_S into $\mathcal{P}_\Sigma(\mathsf{Sm}_S)$, the category of presheaves that convert finite coproducts to finite products. Since Sm_S is an extensive category, we have $\mathcal{P}_\Sigma(\mathsf{Sm}_S) \simeq \mathsf{Shv}_\sqcup(\mathsf{Sm}_S)$, which is a topos.

(ii) **Nisnevich descent.** The presheaves in $\mathcal{P}_\Sigma(\mathsf{Sm}_S)$ do not necessarily satisfy the descent conditions we need. We force Nisnevich covers to become equivalence relations by applying Nisnevich sheafification.

(iii) **$\mathbb{A}^1$-localization.** We force the maps $\mathbb{A}^1 \times_S X \to X$ to become isomorphisms. The result is the category of motivic spaces $\mathsf{H}(S)$.

***Remark* 1.1.3** (Why $\mathcal{P}_\Sigma$?)**.** One could also develop motivic homotopy theory using the full presheaf category $\mathsf{PShv}(\mathsf{Sm}_S)$. The advantage of $\mathcal{P}_\Sigma(\mathsf{Sm}_S)$ is threefold:

(i) It is already a topos: since Sm_S is extensive, $\mathcal{P}_\Sigma(\mathsf{Sm}_S) \simeq \mathsf{Shv}_\sqcup(\mathsf{Sm}_S)$ [BH21, Lemma 2.4].

(ii) The Yoneda embedding $\mathsf{Sm}_S \hookrightarrow \mathcal{P}_\Sigma(\mathsf{Sm}_S)$ preserves finite coproducts, and every object is a sifted colimit of representables.

(iii) In the construction of norm functors, the universal property of $\mathcal{P}_\Sigma$—as the free sifted cocompletion—provides crucial existence and uniqueness guarantees; see [BH21, §3].

Since finite coproduct decompositions are Nisnevich covers, there is a natural inclusion

$$\mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_S) \subset \mathcal{P}_\Sigma(\mathsf{Sm}_S) \subset \mathsf{PShv}(\mathsf{Sm}_S).$$

We write $\mathrm{L}_{\mathrm{Nis}} : \mathcal{P}_\Sigma(\mathsf{Sm}_S) \to \mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_S)$ for the Nisnevich sheafification functor.

1.2. The Nisnevich Topology

In this section we introduce the **Nisnevich topology**, also known as the completely decomposed topology. In motivic homotopy theory, the choice of Nisnevich topology is dictated by two complementary optimality conditions: it is the coarsest topology for which the purity theorem holds, and the finest topology for which algebraic K-theory is representable.

1.2.1. Comparison of three topologies

For a quasi-compact quasi-separated scheme S, let Sm_S be the category of smooth S-schemes of finite presentation. There are three commonly used Grothendieck topologies on Sm_S, listed from coarsest to finest:

$$\mathrm{Zar} \subset \mathrm{Nis} \subset \text{ét}.$$

- **Zariski topology:** Covers consist of open immersions. This provides the weakest form of localization.

- **Étale topology:** Covers consist of étale morphisms (required to be jointly surjective). This allows finite separable field extensions.
- **Nisnevich topology:** Covers consist of étale morphisms, but with the additional requirement that *no residue field extensions occur*. This lies between the other two.

The intuition for the Nisnevich topology can be understood from the functor-of-points perspective. For a smooth scheme X over a field k, an étale cover $\{U_i \to X\}$ does not necessarily satisfy $\coprod_i U_i(k) \twoheadrightarrow X(k)$: the preimage of some k-rational points might only exist over a finite separable extension L/k. A Nisnevich cover precisely excludes this phenomenon—every k-rational point lifts without field extension.

***Remark* 1.2.1** (Why Nisnevich?). The Nisnevich topology is characterized by two complementary optimality conditions:

- **It is the coarsest topology for which the purity theorem holds.** The purity theorem asserts that for a closed immersion $Z \hookrightarrow X$ in Sm_S with normal bundle ν_Z, there is an $\mathbb{A}^1$-local isomorphism $X/(X-Z) \simeq \mathrm{Th}(\nu_Z)$. This is the source of Gysin sequences and purity isomorphisms. The Zariski topology is too coarse for this theorem to hold.
- **It is the finest topology for which algebraic K-theory is representable.** Non-connective K-theory $\mathrm{K}\colon \mathsf{Sm}_S^{\mathrm{op}} \to \mathsf{Sp}$ satisfies Nisnevich descent but not étale descent. Using the étale topology would destroy representability.

1.2.2. Nisnevich covers

Definition 1.2.2 (Nisnevich cover). Let $X \in \mathsf{Sm}_S$. A family of étale morphisms $(f_i\colon U_i \to X)_{i\in I}$ is called a **Nisnevich cover** if for every point $x \in X$, there exists $i \in I$ and $u \in U_i$ such that $f_i(u) = x$ and f_i induces an isomorphism of residue fields $\kappa(u) \xrightarrow{\sim} \kappa(x)$.

***Remark* 1.2.3.** The term "completely decomposed" originates from arithmetic: for a scheme over the ring of integers O_K of a number field K, a prime ideal $\mathfrak{p}$ is "completely decomposed" in an étale cover precisely when its residue field does not extend.

The following theorem provides equivalent characterizations of Nisnevich covers.

Theorem 1.2.4 (Equivalent characterizations). *Let X be a quasi-compact quasi-separated scheme, and let $(f_i\colon U_i \to X)_{i\in I}$ be a family of étale morphisms. The following are equivalent:*

(i) ***(Rational points)*** *For every point $x \in X$, there exists $i \in I$ and $u \in U_i$ with $f_i(u) = x$ and $\kappa(u) \xrightarrow{\sim} \kappa(x)$.*

(ii) ***(Finite stratification)*** *There exists a chain of finitely presented closed subschemes*

$$\varnothing = Z_0 \subset Z_1 \subset \cdots \subset Z_n = X$$

such that for each $j = 1, \dots, n$, some f_i admits a section over $Y_j = Z_j \setminus Z_{j-1}$.

(iii) ***(Refinement by Nisnevich squares)*** *There exists a stratification as above and a family of finitely presented étale morphisms $(g_j\colon V_j \to X)_{j=1}^n$ refining $(f_i)_{i\in I}$, such that g_j restricts to an isomorphism over $Y_j = Z_j \setminus Z_{j-1}$.*

Proof. The implications (iii) ⇒ (ii) ⇒ (i) are immediate. We prove (i) ⇒ (iii).

Replacing each U_i by an affine open cover, we may assume all U_i are quasi-compact and separated, hence all f_i are of finite presentation. Consider the collection of closed subschemes $Y \subseteq X$ for which (iii) fails. We show this collection is empty using Zorn's lemma.

Let $(Y_\alpha)_{\alpha\in A}$ be a chain, and set $Y = \bigcap_\alpha Y_\alpha$. If (iii) holds for Y, then by finite presentation of the relevant morphisms, passing to the limit shows (iii) holds for some Y_α. By Zorn's lemma, if (iii) fails, there exists a minimal closed subscheme $Y \subseteq X$ for which it fails. Replacing X by Y, we may assume (iii) fails for X but holds for all proper closed subschemes.

Clearly $X \neq \varnothing$. Take a generic point $x \in X$ (i.e., one with no non-trivial generizations). By (i), for some i there exists a morphism $\mathrm{Spec}(\kappa(x)) \to U_i$ lifting $\mathrm{Spec}(\kappa(x)) \to X$. Since $O_{X,x}^{\mathrm{red}} = \kappa(x)$, the nilpotent lifting property for étale morphisms yields a lift $\mathrm{Spec}(O_{X,x}) \to U_i$. Passing to a limit, there exists a quasi-compact open neighborhood $V \subseteq X$ of x and a section $s\colon V \to U_i$.

Take a finitely presented closed subscheme $Z \subset X$ with $V = X \setminus Z$. By minimality, (iii) holds for Z: there exists a stratification $\varnothing = Z_0 \subset Z_1 \subset \cdots \subset Z_n = Z$ and a refining family $(h_j \colon W_j \to Z)_{j=1}^n$ satisfying the required conditions. For each j, let $V_j \subseteq U_{i_j}$ be a finitely presented open subscheme such that $g_j \colon V_j \to X$ is étale and restricts to an isomorphism over Y_j. Extending the chain to $\varnothing = Z_0 \subset \cdots \subset Z_n \subset Z_{n+1} = X$ with $V_{n+1} = V$ and $g_{n+1} = f_i \circ s$ yields a realization of (iii) for X, a contradiction. □

1.2.3. Nisnevich squares

The key feature of the Nisnevich topology in practice is that it can be generated by a particularly simple class of covers called "Nisnevich squares."

Definition 1.2.5 (Nisnevich square). A pullback square in Sm_S

$$\begin{tikzcd} U \times_X V \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & V \arrow[d, "p"] \\ U \arrow[r, hook, "j"'] & X \end{tikzcd}$$

is called a **Nisnevich square** if

- j is an open immersion,
- p is étale,
- p induces an isomorphism $p^{-1}(X \setminus U)_{\mathrm{red}} \xrightarrow{\sim} (X \setminus U)_{\mathrm{red}}$.

From the functor-of-points perspective, a Nisnevich square says that the R-points of X split into two types: those landing in the open subfunctor U, and those in the closed complement $X \setminus U$. For the latter, the étale map p provides a cover without changing residue fields.

Lemma 1.2.6. *The morphism family $\{j \colon U \hookrightarrow X,\ p \colon V \to X\}$ forms a Nisnevich cover of X.*

Proof. Let $x \in X$. If $x \in U$, then j satisfies the condition of Definition 1.2.2 at x. If $x \in X \setminus U$, then by assumption there exists $v \in p^{-1}(x)$ with $\kappa(v) \simeq \kappa(x)$. □

More importantly, Nisnevich squares *generate* the Nisnevich topology:

Proposition 1.2.7 (Brown–Gersten property). *Let S be a quasi-compact quasi-separated scheme. A presheaf $\mathcal{F}$ on Sm_S is a Nisnevich sheaf if and only if $\mathcal{F}(\varnothing)$ is terminal and for every Nisnevich square, the natural map*

$$\mathcal{F}(X) \xrightarrow{\sim} \mathcal{F}(V) \times_{\mathcal{F}(U \times_X V)} \mathcal{F}(U)$$

is an isomorphism. In other words, $\mathcal{F}$ sends Nisnevich squares to pullback squares.

Proof. ($\Rightarrow$) If $\mathcal{F}$ is a Nisnevich sheaf, Lemma 1.2.6 shows that $\{U \hookrightarrow X, V \to X\}$ is a Nisnevich cover. The sheaf condition (local sections agreeing on overlaps can be glued) directly yields the pullback condition. The condition $\mathcal{F}(\varnothing) = *$ follows from the empty cover of $\varnothing$.

($\Leftarrow$) Assume $\mathcal{F}$ sends Nisnevich squares to pullbacks and $\mathcal{F}(\varnothing) = *$. We must show $\mathcal{F}$ satisfies the sheaf condition for any Nisnevich cover of X. By Theorem 1.2.4(iii), there exists a refining family $(g_j \colon V_j \to X)_{j=1}^n$ with a stratification $\varnothing = Z_0 \subset Z_1 \subset \cdots \subset Z_n = X$ such that g_j is an isomorphism over $Y_j = Z_j \setminus Z_{j-1}$. We proceed by induction on n.

When $n = 0$, we have $X = \varnothing$, and the condition $\mathcal{F}(\varnothing) = *$ gives the result. For $n \geq 1$, consider the bottom stratum Z_1. Since $Z_0 = \varnothing$, $g_1 \colon V_1 \to X$ is an isomorphism over Z_1. Let $U = X \setminus Z_1$, so $j \colon U \hookrightarrow X$ is an open immersion with $X \setminus U = Z_1$. This yields a Nisnevich square, and by assumption $\mathcal{F}(X) \simeq \mathcal{F}(V_1) \times_{\mathcal{F}(U \times_X V_1)} \mathcal{F}(U)$.

Now U inherits a strictly shorter stratification of length $n-1$, and by the inductive hypothesis, $\mathcal{F}$ satisfies the sheaf condition on U (and similarly on $U \times_X V_1$). By transitivity of pullbacks, $\mathcal{F}$ satisfies the sheaf condition on X. □

1.2.4. Examples

Example 1.2.8 (Zariski covers are Nisnevich covers). Open immersions induce identity maps on residue fields, so Zariski covers automatically satisfy Definition 1.2.2. In particular, the standard cover $\{\mathbb{A}^1 \hookrightarrow \mathbb{P}^1,\ \mathbb{A}^1 \hookrightarrow \mathbb{P}^1\}$ of $\mathbb{P}^1$ is a Nisnevich cover, corresponding to the Nisnevich square

$$\begin{tikzcd} \mathbb{G}_m \arrow[r] \arrow[d] & \mathbb{A}^1 \arrow[d] \\ \mathbb{A}^1 \arrow[r] & \mathbb{P}^1. \end{tikzcd}$$

Example 1.2.9 (The m-th power map). Let k be a field with $\mathrm{char}(k) \nmid m$, and let $a \in k \setminus \{0\}$. Consider the morphism

$$\begin{aligned} f\colon (\mathbb{A}^1 \setminus \{a\}) \sqcup (\mathbb{A}^1 \setminus \{0\}) &\longrightarrow \mathbb{A}^1, \\ \text{first component:} \quad x &\longmapsto x, \\ \text{second component:} \quad y &\longmapsto y^m. \end{aligned}$$

The first component is an open immersion; the second is étale when $\mathrm{char}(k) \nmid m$. This f is a Nisnevich cover if and only if a has an m-th root in k.

Example 1.2.10 (Algebraic K-theory). Algebraic K-theory is a Nisnevich sheaf but not an étale sheaf. To verify the Nisnevich sheaf property, by the Brown–Gersten property it suffices to check Nisnevich excision (since $\mathrm{K}(\varnothing) = 0$ is clear). For any Nisnevich square

$$\begin{tikzcd} W \arrow[r] \arrow[d] & V \arrow[d] \\ U \arrow[r, "j"'] & Y, \end{tikzcd}$$

we need to show the induced square of K-theory spectra is a pullback. Since perfect complexes satisfy Nisnevich descent (in fact, flat descent), the Brown–Gersten property gives a pullback square

$$\begin{tikzcd} \mathsf{Perf}(W) & \mathsf{Perf}(V) \arrow[l] \\ \mathsf{Perf}(U) \arrow[u] & \mathsf{Perf}(Y) \arrow[l] \arrow[u] \end{tikzcd}$$

of stable categories. Setting $Z = Y \setminus U = V \setminus W$ and taking horizontal fibers yields a diagram with rows that are Karoubi sequences (using the fully faithfulness of $j_*\colon \mathsf{D}(U) \to \mathsf{D}(Y)$). Since algebraic K-theory sends Karoubi sequences to exact sequences of spectra, the K-theory square is also a pullback.

1.2.5. Milnor excision

In the affine setting, a particularly important class of Nisnevich squares arises:

Definition 1.2.11 (Milnor square). A **Milnor square** for a commutative ring R is a pullback of the form

$$\begin{tikzcd} R \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & R/I \arrow[d] \\ R_f \arrow[r] & R_f/IR_f \end{tikzcd}$$

where $f \in R$, $I \subset R$ is an ideal, and the natural map $R/I \to R_f/IR_f$ is an isomorphism (equivalently, f is a unit in R/I).

From the functor-of-points perspective, a Milnor square is precisely a Nisnevich square for affine schemes: $U = \mathrm{Spec}(R_f) \hookrightarrow \mathrm{Spec}(R) = X$ is the principal open defined by f, and $V = \mathrm{Spec}(R/I) \to X$ is a closed immersion. The condition $R/I \simeq R_f/IR_f$ says that $p^{-1}(X \setminus U) \simeq X \setminus U$.

Proposition 1.2.12 (Milnor excision). *Let $\mathcal{F}$ be a Nisnevich sheaf on Sm_S. Then for every Milnor square, there is a natural isomorphism*

$$\mathcal{F}(\mathrm{Spec}(R)) \xrightarrow{\sim} \mathcal{F}(\mathrm{Spec}(R_f)) \times_{\mathcal{F}(\mathrm{Spec}(R_f/IR_f))} \mathcal{F}(\mathrm{Spec}(R/I)).$$

1.3. The cdh Topology

In the Nisnevich topology (and the étale topology), covering morphisms are required to be étale. Geometrically, étale morphisms are analogous to local homeomorphisms: they preserve dimension and tangent spaces. Consequently, *étale covers cannot resolve singularities*.

Recall the nodal curve $N = \mathrm{Spec}(k[x,y]/(y^2 - x^3 - x^2))$. The normalization map $\varphi\colon \tilde{N} \to N$ is an isomorphism away from the origin, but at the origin it collapses two points to one and is certainly not étale.

If we wish to study singular schemes within a unified framework—reducing their properties to those of smooth schemes—we need a finer topology that treats "blowups" as covers. This is the motivation for the **cdh topology** (complete decomposition with h-descent).

1.3.1. Abstract blowup squares

Definition 1.3.1 (Abstract blowup square). A commutative diagram in Sch_S

$$\begin{tikzcd} E \arrow[r] \arrow[d, "q"'] \arrow[dr, phantom, "\lrcorner", very near start] & \tilde{X} \arrow[d, "p"] \\ Z \arrow[r, hook, "i"'] & X \end{tikzcd}$$

is called an **abstract blowup square** if

- i is a closed immersion,
- p is proper,
- p induces an isomorphism $p^{-1}(X \setminus Z)_{\mathrm{red}} \xrightarrow{\sim} (X \setminus Z)_{\mathrm{red}}$,
- $E = \tilde{X} \times_X Z$ (with reduced structure) is the pullback.

***Remark* 1.3.2** (Comparison with Nisnevich squares). The structure of abstract blowup squares is parallel to Nisnevich squares:

	Nisnevich square	**Abstract blowup square**
Decomposition	$U \hookrightarrow X$	$Z \hookrightarrow X$
Covering morphism	$V \to X$ étale	$\tilde{X} \to X$ proper
Isomorphism locus	$V\vert_{X\setminus U} \xrightarrow{\sim} X \setminus U$	$\tilde{X}\vert_{X\setminus Z} \xrightarrow{\sim} X \setminus Z$
Typical source	Étale neighborhoods	Blowups, normalizations

Both decompose X into a "known part" and a "part needing coverage," then use an appropriate morphism to handle the latter.

Example 1.3.3 (cdh cover of the nodal curve). Consider the nodal curve $N = \mathrm{Spec}(k[x,y]/(y^2 - x^3 - x^2))$:

$$\begin{tikzcd} E \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \tilde{N} \simeq \mathbb{A}^1 \arrow[d, "p"] \\ Z = \{0\} \arrow[r, hook] & N \end{tikzcd}$$

Here $p\colon \tilde{N} \to N$ is the normalization (a finite, hence proper, morphism). When $\mathrm{char}(k) \neq 2$, the exceptional fiber is $E = \mathrm{Spec}(k[c]/(c^2-1)) \simeq \mathrm{Spec}(k) \sqcup \mathrm{Spec}(k)$, corresponding to the two tangent directions at the node.

For any cdh sheaf $\mathcal{F}$, this gives

$$\mathcal{F}(N) \simeq \mathcal{F}(\mathbb{A}^1) \times_{\mathcal{F}(\{c=1\}\sqcup\{c=-1\})} \mathcal{F}(\{0\}).$$

That is, invariants on the nodal curve are completely determined by invariants on the affine line, the singular point, and gluing data on the exceptional fiber.

1.3.2. Definition of the cdh topology

Definition 1.3.4 (cdh topology). Let X be a quasi-compact quasi-separated scheme. The **cdh topology** on $\mathsf{Sch}_X^{\mathrm{qcqs}}$ is the Grothendieck topology generated by:

(i) Nisnevich squares (Definition 1.2.5),

(ii) Abstract blowup squares (Definition 1.3.1).

A presheaf $\mathcal{F}$ on $\mathsf{Sch}_X^{\mathrm{qcqs}}$ is a **cdh sheaf** if it sends both types of squares to pullback squares. In other words, $\mathcal{F}$ is a cdh sheaf if and only if it satisfies both Nisnevich excision and proper excision.

1.3.3. Extension from smooth to singular schemes

The deepest application of the cdh topology is systematic extension of cohomology theories from smooth schemes Sm_S to general schemes $\mathsf{Sch}_S^{\mathrm{qcqs}}$.

Construction 1.3.5 (Left Kan extension + cdh sheafification). Let $\mathcal{F} : \mathsf{Sm}_S^{\mathrm{op}} \to \mathsf{Sp}$ be a presheaf of spectra on smooth S-schemes. To extend $\mathcal{F}$ to all quasi-compact quasi-separated S-schemes, we proceed in two steps:

(i) **Left Kan extension.** Extend along the inclusion $\mathsf{Sm}_S \hookrightarrow \mathsf{Sch}_S^{\mathrm{qcqs}}$:

$$\begin{tikzcd} \mathsf{Sm}_S^{\mathrm{op}} \arrow[r, "\mathcal{F}"] \arrow[d, hook] & \mathsf{Sp} \\ (\mathsf{Sch}_S^{\mathrm{qcqs}})^{\mathrm{op}} \arrow[ur, "\mathrm{Lan}^{\mathrm{sm}}\mathcal{F}"'] & \end{tikzcd}$$

Explicitly, for a general scheme Y,

$$(\mathrm{Lan}^{\mathrm{sm}}\mathcal{F})(Y) = \mathrm{colim}_{(U\to Y)\in((\mathsf{Sm}_S)_{/Y})^{\mathrm{op}}} \mathcal{F}(U).$$

(ii) **cdh sheafification.** Apply cdh sheafification:

$$\mathcal{F}_{\mathrm{cdh}} \coloneqq \mathrm{L}_{\mathrm{cdh}}(\mathrm{Lan}^{\mathrm{sm}}\mathcal{F}).$$

When $\mathcal{F}$ is already a Nisnevich sheaf on smooth schemes, $\mathcal{F}_{\mathrm{cdh}}$ agrees with $\mathcal{F}$ on smooth schemes but acquires "correct" values on singular schemes via cdh descent.

1.4. $\mathbb{A}^1$-Localization

Write $\mathsf{PShv}_{\mathbb{A}^1}(\mathsf{Sm}_S) \subset \mathsf{PShv}(\mathsf{Sm}_S)$ for the full subcategory of $\mathbb{A}^1$-invariant presheaves, i.e., those $\mathcal{F}$ satisfying $\mathcal{F}(U) \xrightarrow{\sim} \mathcal{F}(U \times_S \mathbb{A}^1)$ for all $U \in \mathsf{Sm}_S$. The inclusion admits a left adjoint

$$\mathrm{L}_{\mathbb{A}^1} : \mathsf{PShv}(\mathsf{Sm}_S) \to \mathsf{PShv}_{\mathbb{A}^1}(\mathsf{Sm}_S),$$

called the **$\mathbb{A}^1$-localization functor**.

Definition 1.4.1 ($\mathbb{A}^1$-local isomorphism and $\mathbb{A}^1$-homotopy). Let $f : \mathcal{F} \to \mathcal{G}$ be a morphism in $\mathsf{PShv}(\mathsf{Sm}_S)$.

- f is an $\mathbb{A}^1$**-local isomorphism** if $\mathrm{L}_{\mathbb{A}^1}(f)$ is an isomorphism.
- Two maps $f, g\colon \mathcal{F} \to \mathcal{G}$ are $\mathbb{A}^1$**-homotopic** if there exists $H\colon \mathcal{F} \times \mathbb{A}^1 \to \mathcal{G}$ with $H \circ i_0 \simeq f$ and $H \circ i_1 \simeq g$, where $i_s\colon * \to \mathbb{A}^1$ is the inclusion of the point s.
- f is an $\mathbb{A}^1$**-homotopy equivalence** if there exists $g\colon \mathcal{G} \to \mathcal{F}$ such that $f \circ g$ and $g \circ f$ are $\mathbb{A}^1$-homotopic to the respective identities.

Proposition 1.4.2. *Every $\mathbb{A}^1$-homotopy equivalence is an $\mathbb{A}^1$-local isomorphism. In particular, for any presheaf $\mathcal{F}$, the projection $p\colon \mathcal{F} \times \mathbb{A}^1 \to \mathcal{F}$ is an $\mathbb{A}^1$-local isomorphism.*

Proof. We first show $p\colon \mathcal{F} \times \mathbb{A}^1 \to \mathcal{F}$ is an $\mathbb{A}^1$-homotopy equivalence. Since $\mathsf{PShv}(\mathsf{Sm}_S)$ is a topos with universal colimits, it suffices to check this for representable presheaves. For $X \in \mathsf{Sm}_S$, the zero section $s\colon X \to X \times \mathbb{A}^1$ satisfies $p \circ s = \mathrm{id}_X$. The reverse composite $s \circ p$ is $\mathbb{A}^1$-homotopic to $\mathrm{id}_{X\times\mathbb{A}^1}$ via

$$(X \times \mathbb{A}^1) \times \mathbb{A}^1 \xrightarrow{\mathrm{id}_X \times \mu} X \times \mathbb{A}^1,$$

where $\mu\colon \mathbb{A}^1 \times \mathbb{A}^1 \to \mathbb{A}^1$ is the ring multiplication.

Now suppose $a, b\colon \mathcal{F} \to \mathcal{G}$ are $\mathbb{A}^1$-homotopic via H. Since $\mathrm{L}_{\mathbb{A}^1}(p)$ is an isomorphism, and $p \circ i_0 = p \circ i_1 = \mathrm{id}$, applying $\mathrm{L}_{\mathbb{A}^1}$ gives $\mathrm{L}_{\mathbb{A}^1}(i_0) = \mathrm{L}_{\mathbb{A}^1}(i_1)$. Hence

$$\mathrm{L}_{\mathbb{A}^1}(a) = \mathrm{L}_{\mathbb{A}^1}(H) \circ \mathrm{L}_{\mathbb{A}^1}(i_0) = \mathrm{L}_{\mathbb{A}^1}(H) \circ \mathrm{L}_{\mathbb{A}^1}(i_1) = \mathrm{L}_{\mathbb{A}^1}(b).$$

The result follows from the definition of homotopy equivalence. □

1.4.1. Explicit formula for $\mathrm{L}_{\mathbb{A}^1}$

The $\mathbb{A}^1$-localization functor can be computed via an explicit "singular functor."

Definition 1.4.3 (Algebraic simplex)**.** The standard n-simplex over S is

$$\Delta^n_S := \mathrm{Spec}\big(\mathcal{O}_S[T_0, \dots, T_n]/(T_0 + \cdots + T_n - 1)\big) \subset \mathbb{A}^{n+1}_S.$$

Face and degeneracy maps are defined analogously to the topological case, yielding a cosimplicial scheme $\Delta^\bullet_S\colon \mathbf{\Delta} \to \mathsf{Sch}_S$. Note that $\Delta^n_S \simeq \mathbb{A}^n_S$ as S-schemes (but not as cosimplicial objects).

Theorem 1.4.4 (Explicit formula for $\mathbb{A}^1$-localization)**.** *For $\mathcal{F} \in \mathsf{PShv}(\mathsf{Sm}_S)$ and $X \in \mathsf{Sm}_S$,*

$$(\mathrm{L}_{\mathbb{A}^1}\mathcal{F})(X) \simeq \mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathcal{F}(X \times \Delta^n_S).$$

Proof. Write $\mathcal{F}'(X) := \mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathcal{F}(X \times \Delta^n_S)$.

$\mathbb{A}^1$**-invariance.** The projection $p\colon X\times\mathbb{A}^1 \to X$ and zero section s induce morphisms between simplicial objects. We have $s^* \circ p^* = \mathrm{id}$, while $p^* \circ s^*$ admits an $\mathbb{A}^1$-homotopy via the multiplication $\mu\colon \mathbb{A}^1 \times \mathbb{A}^1 \to \mathbb{A}^1$. The key observation is that $\Delta^1_S \simeq \mathbb{A}^1_S$, so this algebraic homotopy is equivalent to a simplicial homotopy. Taking colimits over $\mathbf{\Delta}^{\mathrm{op}}$, simplicial homotopies become isomorphisms, yielding $p^*\colon \mathcal{F}'(X) \xrightarrow{\sim} \mathcal{F}'(X \times \mathbb{A}^1)$.

Universal property. Let $\mathcal{G}$ be $\mathbb{A}^1$-invariant. Since $\Delta^n_S \simeq \mathbb{A}^n_S$, the simplicial object $\mathcal{G}(X \times \Delta^\bullet_S)$ is constant, so $\mathrm{colim}_{\mathbf{\Delta}^{\mathrm{op}}} \mathcal{G}(X \times \Delta^\bullet_S) \simeq \mathcal{G}(X)$. Hence

$$\mathrm{Hom}(\mathcal{F}', \mathcal{G}) \simeq \lim_{\mathbf{\Delta}} \mathrm{Hom}(\mathcal{F}(- \times \Delta^n_S), \mathcal{G}) \simeq \mathrm{Hom}(\mathcal{F}, \lim_{\mathbf{\Delta}} \mathcal{G}(- \times \Delta^n_S)) \simeq \mathrm{Hom}(\mathcal{F}, \mathcal{G}).$$ □

***Remark* 1.4.5** (Geometric meaning)**.** Theorem 1.4.4 is the algebro-geometric analogue of the classical singular functor Sing in topology. The simplicial anima $\mathcal{F}(X\times\Delta^\bullet_S)$ encodes algebraic homotopy information: 0-simplices are sections, 1-simplices parametrize $\mathbb{A}^1$-paths, 2-simplices parametrize higher homotopies, and so on. Taking the colimit over $\mathbf{\Delta}^{\mathrm{op}}$ quotients out all $\mathbb{A}^1$-deformations.

However, $\mathrm{L}_{\mathbb{A}^1}$ does not generally preserve Nisnevich sheaves, so the motivic localization $\mathrm{L}_{\mathrm{mot}}$ cannot simply be written as $\mathrm{L}_{\mathbb{A}^1} \circ \mathrm{L}_{\mathrm{Nis}}$; it requires alternating iterations (see Remark 1.5.3).

1.4.2. Interval objects

We have encountered three constructions related to $\mathbb{A}^1$: $\mathbb{A}^1$-localization, localization at $\mathbb{A}^1$-homotopy equivalences, and the $\mathrm{L}_{\mathbb{A}^1}$ construction. A natural question is: do they coincide? The answer depends on a key algebraic property of $\mathbb{A}^1$: it is an **interval object**.

This subsection follows Hoyois [Hoy], clarifying that the good properties of $\mathbb{A}^1$-homotopy theory are formal consequences of the interval object structure.

Definition 1.4.6 (Bipointed object)**.** Let $\mathcal{C}$ be a category with finite products. A **bipointed object** is an object $I \in \mathcal{C}$ equipped with two global sections $0, 1 \colon * \rightrightarrows I$.

Definition 1.4.7 (Three homotopy categories)**.** Let $\mathcal{C}$ be a category with finite products and I a bipointed object.

(i) **I-localization** (nullification): formally invert all projections $X \times I \to X$:

$$\mathcal{C}/I \;:=\; \mathcal{C}[\{X \times I \to X\}^{-1}].$$

(ii) **Localization at I-homotopy equivalences**: formally invert only I-homotopy equivalences:

$$\mathcal{C}[\mathrm{hoeq}_I^{-1}].$$

(iii) **I-homotopy category**: via simplicial enrichment from the cosimplicial extension:

$$\mathrm{Ho}(\mathcal{C}, I), \quad \mathrm{Hom}_{\mathrm{Ho}(\mathcal{C},I)}(X, Y) = \mathrm{colim}_{\mathbf{\Delta}^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{C}}(X \times I^n, Y).$$

In general, there are canonical functors

$$\mathcal{C}[\mathrm{hoeq}_I^{-1}] \longrightarrow \mathrm{Ho}(\mathcal{C}, I) \longrightarrow \mathcal{C}/I,$$

but neither is generally an isomorphism.

Definition 1.4.8 (Interval object)**.** A bipointed object I in $\mathcal{C}$ is an **interval object** if:

(i) $X \times I$ exists for all $X \in \mathcal{C}$;

(ii) I is I-contractible, i.e., $I \to *$ is an I-homotopy equivalence.

Example 1.4.9. (i) The unit interval $[0, 1] \in \mathsf{Top}$ with multiplication $\mu(x, y) = xy$ (unit 1, absorbing element 0).

(ii) The affine line $\mathbb{A}^1 \in \mathsf{Sm}_S$ with ring multiplication $\mu(x, y) = xy$.

(iii) The real line $\mathbb{R}$ and complex plane $\mathbb{C}$ in Top, both with multiplication as μ.

The key property of interval objects is that they unify the three homotopy categories:

Theorem 1.4.10 (Morel–Voevodsky [MV99])**.** *Let $\mathcal{C}$ be a category and I an interval object in $\mathcal{C}$. Then*

$$\mathcal{C}[\mathrm{hoeq}_I^{-1}] \xrightarrow{\sim} \mathrm{Ho}(\mathcal{C}, I) \xrightarrow{\sim} \mathcal{C}/I.$$

Proof sketch. The proof has two parts.

Left isomorphism. Since I is I-contractible, there is a zigzag of I-homotopies connecting id_I to the constant map $0 \colon I \to * \xrightarrow{0} I$. Taking products with X shows that $(\mathrm{id}_X, 0) \colon X \to X \times I$ is an I-homotopy inverse to pr_1, so every projection $X \times I \to X$ is an I-homotopy equivalence.

Right isomorphism. One shows that the I-homotopy from id_I to 0 induces a simplicial homotopy between the corresponding endomorphisms of $I \times I^\bullet$. Taking colimits over $\mathbf{\Delta}^{\mathrm{op}}$, this becomes an isomorphism, which implies the mapping anima in $\mathrm{Ho}(\mathcal{C}, I)$ agree with those in $\mathcal{C}/I$. □

Corollary 1.4.11. *The affine line $\mathbb{A}^1$ with sections $0, 1$ and multiplication $\mu(x, y) = xy$ is an interval object in Sm_S (and in $\mathcal{P}_\Sigma(\mathsf{Sm}_S)$). Thus for $\mathcal{C} = \mathcal{P}_\Sigma(\mathsf{Sm}_S)$, the following give the same result:*

(i) $\mathbb{A}^1$-localization: inverting all $\mathcal{F} \times \mathbb{A}^1 \to \mathcal{F}$;

(ii) Localization at $\mathbb{A}^1$-homotopy equivalences;

(iii) The $\mathrm{L}_{\mathbb{A}^1}$ construction.

***Remark* 1.4.12.** The multiplication $\mu(x, y) = xy$ on $\mathbb{A}^1$ provides an $\mathbb{A}^1$-homotopy from $\mathrm{id}_{\mathbb{A}^1}$ (at $y = 1$) to the constant map 0 (at $y = 0$). This is why we take 1 as the basepoint rather than 0: the unit element of the multiplication is what makes $\mathbb{A}^1$ an interval object.

1.5. Motivic Spaces

We now construct the motivic homotopy category. Let S be a quasi-compact quasi-separated scheme. Since Sm_S is extensive, $\mathcal{P}_\Sigma(\mathsf{Sm}_S) \simeq \mathsf{Shv}_\sqcup(\mathsf{Sm}_S)$. Since finite coproduct decompositions are Nisnevich covers, we have

$$\mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_S) \subset \mathcal{P}_\Sigma(\mathsf{Sm}_S).$$

Definition 1.5.1 (Motivic spaces)**.** Let S be a quasi-compact quasi-separated scheme. The category of **motivic spaces** over S is

$$\mathsf{H}(S) \coloneqq \mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_S) \cap \mathsf{PShv}_{\mathbb{A}^1}(\mathsf{Sm}_S) \subset \mathsf{PShv}(\mathsf{Sm}_S),$$

i.e., presheaves (valued in An) satisfying both Nisnevich descent and $\mathbb{A}^1$-invariance. The motivic localization functor is denoted $\mathrm{L}_{\mathrm{mot}} \colon \mathsf{PShv}(\mathsf{Sm}_S) \to \mathsf{H}(S)$.

A morphism f in $\mathsf{PShv}(\mathsf{Sm}_S)$ is a **motivic equivalence** if $\mathrm{L}_{\mathrm{mot}}(f)$ is an isomorphism.

Fact 1.5.2. *(i) $\mathrm{L}_{\mathrm{Nis}}$ is left exact. Both $\mathrm{L}_{\mathbb{A}^1}$ and $\mathrm{L}_{\mathrm{mot}}$ preserve finite products [Hoy14, Proposition C.6].*

(ii) Motivic equivalences form the smallest strongly saturated class in $\mathsf{PShv}(\mathsf{Sm}_S)$ containing both Nisnevich local isomorphisms and $\mathbb{A}^1$-local isomorphisms.

***Remark* 1.5.3** (Non-commutativity of localizations)**.** In general, $\mathrm{L}_{\mathrm{mot}} \neq \mathrm{L}_{\mathbb{A}^1} \circ \mathrm{L}_{\mathrm{Nis}}$ and $\mathrm{L}_{\mathrm{mot}} \neq \mathrm{L}_{\mathrm{Nis}} \circ \mathrm{L}_{\mathbb{A}^1}$: $\mathrm{L}_{\mathbb{A}^1}$ can send Nisnevich sheaves to non-sheaves, and $\mathrm{L}_{\mathrm{Nis}}$ can destroy $\mathbb{A}^1$-invariance. But $\mathsf{H}(S)$ is defined as an intersection, independent of the order. The actual computation is $\mathrm{L}_{\mathrm{mot}} \simeq (\mathrm{L}_{\mathrm{Nis}} \circ \mathrm{L}_{\mathbb{A}^1})^{\circ\mathbb{N}}$, i.e., alternating iteration until stabilization.

1.5.1. $\mathbb{A}^1$-rigidity and $\mathbb{A}^1$-connected components

Definition 1.5.4 ($\mathbb{A}^1$-rigidity)**.** An S-scheme X is **$\mathbb{A}^1$-rigid** if $\mathrm{L}_{\mathbb{A}^1}(X) \simeq X$ in $\mathcal{P}_\Sigma(\mathsf{Sm}_S)$, i.e., $X(U \times_S \mathbb{A}^1) \simeq X(U)$ for all $U \in \mathsf{Sm}_S$.

Proposition 1.5.5. *Let k be a field. The following k-schemes are $\mathbb{A}^1$-rigid:*

(i) The multiplicative group $\mathbb{G}_m$;

(ii) Smooth projective curves of genus ≥ 1;

(iii) Abelian varieties.

More generally, if S is a reduced scheme of finite Krull dimension, then $\mathbb{G}_m$ is $\mathbb{A}^1$-rigid in $\mathsf{H}(S)$.

Proof. For $\mathbb{G}_m$: we have $\mathbb{G}_m(U \times \mathbb{A}^1) = \mathcal{O}(U \times \mathbb{A}^1)^\times = \mathcal{O}(U)^\times = \mathbb{G}_m(U)$, where the second equality uses $\mathcal{O}(U)[t]^\times = \mathcal{O}(U)^\times$ when U is reduced. For smooth projective curves and abelian varieties, this follows from the fact that there are no non-constant morphisms from $\mathbb{A}^1$ to a proper variety. □

1.5.2. Motivic spaces and K-theory

Connective K-theory $\mathrm{k} \colon \mathsf{Sm}_S^{\mathrm{op}} \to \mathsf{An}$ satisfies Nisnevich descent (see Example 1.2.10) and is $\mathbb{A}^1$-invariant when the base scheme is regular, hence defines an object of $\mathsf{H}(S)$.

Theorem 1.5.6 (Representability of K-theory)**.** *Let S be a regular Noetherian scheme of finite Krull dimension. Then K-theory is represented in $\mathsf{H}(S)$ by $\mathbb{Z} \times \mathrm{BGL}_\infty$:*

$$\mathrm{L}_{\mathrm{mot}}(\mathbb{Z} \times \mathrm{BGL}_\infty) \simeq \mathrm{k}.$$

In particular, for affine $U \in \mathsf{Sm}_S$, there is a bijection $[U, \mathbb{Z} \times \mathrm{BGL}_\infty]_{\mathsf{H}(S)} \simeq \mathrm{K}_0(U)$.

***Remark* 1.5.7** (Affine representability)**.** Although Vect_r is not $\mathbb{A}^1$-invariant on all smooth schemes (Remark 1.1.2), applying motivic localization still gives correct values on smooth affine schemes. Asok–Hoyois–Wendt [AHW17] proved: if X is a presheaf on Sm_S such that $\pi_0(X)$ is $\mathbb{A}^1$-invariant on affine schemes and X satisfies affine Nisnevich excision, then $\pi_0(X)(U) \xrightarrow{\sim} [U, X]_{\mathsf{H}(S)}$ for all smooth affine U. Taking $X = \mathrm{BGL}_r$ yields $\pi_0\mathsf{Vect}_r(U) \simeq [U, \mathrm{BGL}_r]_{\mathsf{H}(S)}$ for affine U.

***Remark* 1.5.8** (The role of regularity). Nisnevich descent holds without regularity assumptions (Example 1.2.10), but $\mathbb{A}^1$-invariance requires regularity. For regular Noetherian X, Quillen proved $\mathrm{K}_n(X) \xrightarrow{\sim} \mathrm{K}_n(X \times \mathbb{A}^1)$ for all $n \geq 0$. For singular schemes, this fails: for the dual numbers $R = k[\varepsilon]/(\varepsilon^2)$,

$$\mathrm{K}_1(k[\varepsilon]) \not\simeq \mathrm{K}_1(k[\varepsilon][t]),$$

since $k[\varepsilon][t]$ admits automorphisms (e.g., $x \mapsto x + \varepsilon t$) not pulled back from $k[\varepsilon]$.

To obtain $\mathbb{A}^1$-invariance on non-regular schemes, Weibel introduced **homotopy K-theory** KH, defined as the $\mathbb{A}^1$-invariantization of K-theory:

$$\mathrm{KH}(X) := \mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathrm{K}(X \times \Delta^n).$$

This is precisely the $\mathrm{L}_{\mathbb{A}^1}$ construction applied to K.

1.6. Pointed Motivic Spaces

1.6.1. Basic constructions

Definition 1.6.1 (Pointed motivic spaces). The category of **pointed motivic spaces** over S, denoted $\mathsf{H}_*(S)$, is the category of pointed objects in $\mathsf{H}(S)$.

The functor $(-)_+ \colon \mathsf{H}(S) \to \mathsf{H}_*(S)$ adds a disjoint basepoint: $X_+ = X \sqcup *$ with $*$ as basepoint. It is left adjoint to the forgetful functor.

The **smash product** on $\mathsf{H}_*(S)$ is defined as the pushout

$$X \wedge Y := \mathrm{cofib}(X \vee Y \to X \times Y).$$

This gives $\mathsf{H}_*(S)$ a symmetric monoidal structure with unit $S_+ \simeq S \sqcup *$.

We adopt the following basepoint conventions:

- $\mathbb{A}^1$ is pointed at 1;
- $\mathbb{G}_m$ is pointed at 1;
- $\mathbb{P}^1$ is pointed at $\infty = [1 : 0]$.

Definition 1.6.2 (Suspension and loops). For $X \in \mathsf{H}_*(S)$, define

- the **suspension** $\Sigma X := S^1 \wedge X$, where $S^1 = \Delta^1/\partial\Delta^1$ is the simplicial circle (as a constant sheaf);
- the **loop space** $\Omega X := \mathrm{fib}(* \to X)$, the fiber over the basepoint.

These form an adjoint pair $\Sigma \dashv \Omega$.

1.6.2. Bigraded spheres

In $\mathsf{H}_*(S)$, there are two essentially different "circles":

Definition 1.6.3 (Motivic spheres).

- The **simplicial circle** is $S^{1,0} := S^1$ (as a constant sheaf).
- The **Tate circle** is $S^{1,1} := \mathbb{G}_m$ (pointed at 1).
- The **motivic (p, q)-sphere** is

$$S^{p,q} := (S^{1,0})^{\wedge(p-q)} \wedge (S^{1,1})^{\wedge q}, \quad p \geq q \geq 0.$$

The key computation for motivic spheres is:

Proposition 1.6.4 ($\mathbb{P}^1$ as a motivic sphere). *In $\mathsf{H}_*(S)$, there is a motivic equivalence $\mathbb{P}^1 \simeq S^{2,1} = S^1 \wedge \mathbb{G}_m$.*

Proof. Consider the standard Nisnevich square for $\mathbb{P}^1$:

$$\begin{tikzcd} \mathbb{G}_m \arrow[r] \arrow[d] & \mathbb{A}^1 \arrow[d] \\ \mathbb{A}^1 \arrow[r] & \mathbb{P}^1 \end{tikzcd}$$

In $\mathsf{H}_*(S)$, Nisnevich squares are pushout squares (Brown–Gersten property). Since $\mathbb{A}^1 \simeq *$ in $\mathsf{H}(S)$, this pushout degenerates to

$$\mathbb{P}^1 \simeq \mathrm{cofib}(\mathbb{G}_m \to * \sqcup *) \simeq \Sigma \mathbb{G}_m = S^1 \wedge \mathbb{G}_m = S^{2,1}. \qquad \square$$

***Remark* 1.6.5.** This result makes precise the observation that $\mathbb{G}_m = \mathrm{Spec}(\mathbb{Z}[\mathbb{Z}])$ is in some sense dual to the topological circle $S^1 \simeq B\mathbb{Z}$. The motivic sphere $\mathbb{G}_m$ plays the role of $S^{1,1}$, while the simplicial circle S^1 is $S^{1,0}$. The projective line $\mathbb{P}^1 \simeq S^{2,1}$ is their smash product.

This bigraded structure is what makes motivic homotopy theory essentially richer than classical homotopy theory: the weight direction q encodes "Tate twists" specific to algebraic geometry.

Proposition 1.6.6 (Punctured affine spaces). *In $\mathsf{H}_*(S)$, there is a motivic equivalence $\mathbb{A}^n \setminus \{0\} \simeq S^{2n-1,n}$ for $n \geq 1$.*

Proof. We proceed by induction on n. The case $n = 1$ is Proposition 1.6.4 (noting $\mathbb{A}^1 \setminus \{0\} = \mathbb{G}_m \simeq S^{1,1}$). For the inductive step, consider the Nisnevich square

$$\begin{tikzcd} (\mathbb{A}^{n-1} \setminus \{0\}) \times \mathbb{G}_m \arrow[r] \arrow[d] & \mathbb{A}^{n-1} \times \mathbb{G}_m \arrow[d] \\ (\mathbb{A}^{n-1} \setminus \{0\}) \times \mathbb{A}^1 \arrow[r] & \mathbb{A}^n \setminus \{0\} \end{tikzcd}$$

Since $\mathbb{A}^{n-1} \simeq *$, the pushout reduces to

$$\mathbb{A}^n \setminus \{0\} \simeq S^1 \wedge (\mathbb{A}^{n-1} \setminus \{0\}) \wedge \mathbb{G}_m.$$

By the inductive hypothesis, $\mathbb{A}^{n-1} \setminus \{0\} \simeq S^{2n-3,n-1}$, so

$$\mathbb{A}^n \setminus \{0\} \simeq S^{1,0} \wedge S^{2n-3,n-1} \wedge S^{1,1} = S^{2n-1,n}. \qquad \square$$

Corollary 1.6.7 (Thom spaces and motivic spheres). *In $\mathsf{H}_*(S)$,*

$$\mathbb{A}^n/(\mathbb{A}^n \setminus \{0\}) \simeq S^{2n,n} \quad \textit{and} \quad \mathbb{P}^n/\mathbb{P}^{n-1} \simeq S^{2n,n}.$$

Proof. For the first equivalence, the cofiber sequence $\mathbb{A}^n \setminus \{0\} \to \mathbb{A}^n \to \mathbb{A}^n/(\mathbb{A}^n \setminus \{0\})$ together with $\mathbb{A}^n \simeq *$ gives

$$\mathbb{A}^n/(\mathbb{A}^n \setminus \{0\}) \simeq \Sigma(\mathbb{A}^n \setminus \{0\}) \simeq \Sigma S^{2n-1,n} = S^{2n,n}.$$

For the second, use the standard open cover of $\mathbb{P}^n$ and induction. $\square$

1.7. Norms on Motivic Spaces

In this section we briefly describe the norm functors constructed by Bachmann–Hoyois [BH21]. This additional structure will be used later when discussing multiplicative properties of motivic spectra.

1.7.1. Motivation and statement

Let $p\colon T \to S$ be a finite locally free morphism. In classical algebraic geometry, there is the **Weil restriction** $\mathrm{R}_p\colon \mathsf{Sm}_T \to \mathsf{Sm}_S$, the right adjoint to pullback $p^*\colon \mathsf{Sm}_S \to \mathsf{Sm}_T$, characterized by the universal property

$$\mathrm{Hom}_S(Y, \mathrm{R}_p X) \simeq \mathrm{Hom}_T(Y \times_S T, X).$$

In the pointed setting, Weil restriction naturally extends to a symmetric monoidal functor $p_\otimes\colon \mathsf{Sm}_{T,+} \to \mathsf{Sm}_{S,+}$ (with respect to the smash product). The core question of Bachmann–Hoyois is: can this functor be extended to the motivic homotopy category?

Theorem 1.7.1 (Bachmann–Hoyois [BH21, Theorems 3.3, 4.1]). *Let $p\colon T \to S$ be a finite locally free and universally open morphism. Then there exists a symmetric monoidal functor*

$$p_{\otimes}\colon \mathsf{H}_*(T) \to \mathsf{H}_*(S)$$

satisfying:

(i) $p_{\otimes}$ preserves sifted colimits;

(ii) $p_{\otimes}(X_+) \simeq (\mathrm{R}_p X)_+$ for $X \in \mathsf{Sm}_T$;

(iii) if $T = S^{\sqcup n}$ and p is the fold map, then $p_{\otimes}$ is the n-fold smash product;

(iv) if p is furthermore finite étale, then $p_{\otimes}$ extends to a symmetric monoidal functor $p_{\otimes}\colon \mathsf{SH}(T) \to \mathsf{SH}(S)$.

***Remark* 1.7.2** (Key to the construction: universal property of $\mathcal{P}_\Sigma$). The proof of Theorem 1.7.1 relies on the following construction:

$$\mathsf{Sm}_{T,+} \xrightarrow{(1)} \mathcal{P}_\Sigma(\mathsf{Sm}_T)_* \xrightarrow{(2)} \mathsf{H}_*(T) \xrightarrow{(3)} \mathsf{SH}(T).$$

Each step is a universal construction in $\mathrm{CAlg}(\mathsf{Cat}^{\mathrm{sift}})$ (symmetric monoidal categories with sifted colimits). Hence $p_{\otimes}\colon \mathsf{Sm}_{T,+} \to \mathsf{Sm}_{S,+}$ from Weil restriction has at most one sifted-colimit-preserving symmetric monoidal extension at each step.

Specifically:

(i) $p_{\otimes}$ extends unconditionally to $\mathcal{P}_\Sigma$;

(ii) extension to H_* holds iff $p_{\otimes}$ preserves Nisnevich equivalences and $\mathbb{A}^1$-equivalences;

(iii) further extension to SH holds iff the cofiber of $p_{\otimes}(\mathbb{P}^1_{T,+})$ is invertible in $\mathsf{SH}(S)$.

Condition (2) holds for finite locally free universally open p; condition (3) additionally requires p to be finite étale.

***Remark* 1.7.3.** The functor $p_{\otimes}$ preserves neither arbitrary colimits nor arbitrary limits, so it cannot be part of a six-functor formalism.

This completes our treatment of unstable motivic homotopy theory. In the next chapter, we turn to the stable setting: the category $\mathsf{SH}(S)$ of motivic spectra.

Chapter 2.

Stable Motivic Homotopy Theory

2.1. Grothendieck's Vision and the Universal Property of SH

Algebraic geometry abounds with cohomology theories: ℓ-adic cohomology, crystalline cohomology, de Rham cohomology, Betti cohomology, and so on. Grothendieck envisioned that behind all these theories lies a **universal coefficient system** governing all cohomology theories on schemes.

More precisely, he conjectured the existence of an abelian category $\mathsf{Mot}(S)$ and a functor

$$\mathrm{Mot}\colon \mathsf{Sch}_S^{\mathrm{op}} \to \mathsf{Mot}(S)$$

such that for any "cohomology functor" $h^*\colon \mathsf{Sch}_S^{\mathrm{op}} \to \mathcal{A}$, there exists a unique **realization functor** $\bar{h}^*\colon \mathsf{Mot}(S) \to \mathcal{A}$ making the following diagram commute:

$$\begin{tikzcd} \mathsf{Sch}_S^{\mathrm{op}} \arrow[r, "h^*"] \arrow[dr, "\mathrm{Mot}"'] & \mathcal{A} \\ & \mathsf{Mot}(S) \arrow[u, dashed, "\bar{h}^*"'] \end{tikzcd}$$

In other words, $\mathsf{Mot}(S)$ would be the universal target for all cohomology theories. At that time, stable categories had not yet entered the mathematical landscape, so Grothendieck's vision was formulated in the language of abelian categories.

Today, stable categories have largely supplanted classical homological algebra, and "studying cohomology on schemes" is increasingly understood as "studying **six-functor formalisms** on schemes." Grothendieck's quest for universal coefficients thus naturally reformulates as:

Does there exist a universal six-functor formalism?

The answer depends on what axioms we impose. Different choices of axioms yield different "universal" objects. The motivic six-functor formalism SH arises from a specific list of axioms, formulated precisely by Drew–Gallauer [DG22].

Theorem 2.1.1 (Drew–Gallauer [DG22]; cf. [Sch26, Theorem 11.1]). *Let k be a noetherian base ring with $\mathrm{Spec}(k)$ of finite Krull dimension, and let C be the category of separated finite-type k-schemes, equipped with I = open immersions and P = proper morphisms. Consider the category of presentable six-functor formalisms*

$$\mathsf{D}\colon \mathsf{Corr}(C) \longrightarrow \mathsf{Pr}^L_{\mathrm{st}}$$

in which open immersions are cohomologically étale, proper maps are cohomologically proper, and which satisfy:

(i) ***Cohomological smoothness:*** *every smooth morphism is cohomologically smooth in* D*;*

(ii) ***Excision:*** *for any closed immersion $i\colon Z \hookrightarrow X$ with open complement $j\colon U \hookrightarrow X$, the sequence*

$$j_!\mathbf{1} \longrightarrow \mathbf{1} \longrightarrow i_*\mathbf{1}$$

is a fiber sequence in $\mathsf{D}(X)$*;*

(iii) $\mathbb{A}^1$***-homology trivial:*** *for $\pi\colon \mathbb{A}^1 \to \mathrm{Spec}(k)$, the unit map $\mathbf{1} \to \pi_!\pi^!\mathbf{1}$ is an equivalence.*

This category has an initial object, and that initial object is the motivic six-functor formalism SH *of Morel–Voevodsky.*

The proof of [DG22] is constructive, and recovers the construction of SH that we will give in §2.2: enforcing the existence of left adjoints $f_\sharp$ to smooth pullbacks gives the stable presheaf category $\mathsf{PShv}(\mathsf{Sm}_{/X}; \mathsf{Sp})$; excision implies Nisnevich descent and forces passage to $\mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_{/X}; \mathsf{Sp})$; triviality of $\mathbb{A}^1$-homology forces $\mathbb{A}^1$-localization, yielding $\mathsf{Shv}_{\mathrm{Nis},\mathbb{A}^1}(\mathsf{Sm}_{/X}; \mathsf{Sp})$; finally, $\otimes$-inverting the Tate object yields SH. We refer the reader to [Sch26, §11] for a clean exposition of the proof sketch.

***Remark* 2.1.2** (Realizations of motives)**.** The power of Theorem 2.1.1 lies in its corollary: any other six-functor formalism on C satisfying axioms (1)–(3) automatically receives a unique symmetric monoidal functor from SH, called its **realization**. Concrete examples include:

- ℓ-adic étale cohomology $\mathsf{D}_{\text{ét}}(-, \mathbb{Z}_\ell)$, when ℓ is invertible on the base;
- Betti realization $\mathsf{D}(-(\mathbb{C}), \mathbb{Z})$, for $\mathbb{C}$-schemes;
- de Rham realization in characteristic 0.

Each yields a symmetric monoidal functor from SH and identifies the corresponding cohomology theory as a representable spectrum (the ℓ-adic, Betti, or de Rham realization spectrum). This is the modern conceptual explanation for why "every" reasonable cohomology theory of varieties admits motives in the first place.

However, SH is *not* the ultimate answer to Grothendieck's question. The axioms above—especially $\mathbb{A}^1$-invariance—exclude many important cohomology theories. In particular:

- Crystalline cohomology in characteristic p is not $\mathbb{A}^1$-invariant.
- Prismatic cohomology does not factor through SH.
- More generally, p-adic cohomology theories in characteristic p require fundamentally different frameworks.

Thus $\mathsf{SH}(S)$ should be understood as an *approximation* to Grothendieck's vision—the universal coefficient system for $\mathbb{A}^1$-invariant cohomology theories—rather than the definitive solution. The search for a truly universal framework encompassing all cohomology theories, including p-adic ones, remains an active and central problem in arithmetic geometry.

2.1.1. Six-functor formalisms via span categories

Before turning to the construction of $\mathsf{SH}(S)$, we briefly record the modern framework in which six-functor formalisms are now formulated, since it will clarify what we are ultimately aiming at.

Classically, a six-functor formalism on a category C of geometric objects is a packet of data $(f^*, f_*, f_!, f^!, \otimes, \underline{\mathrm{Hom}})$ subject to an intricate web of adjunctions, base change isomorphisms, and projection formula. Even writing down this data in a fully coherent categorical sense is a non-trivial task.

A far more economical description was initiated by Liu–Zheng [LZ24] and Gaitsgory–Rozenblyum [**?**], and brought to its present form by Mann [Man22] and Scholze [**?**]. Given a wide subcategory $E \subset C$ closed under composition and base change, one forms the **span category** $\mathsf{Span}(C, E)$: its objects are those of C, and its morphisms $X \to Y$ are spans

$$X \longleftarrow Z \xrightarrow{e} Y, \qquad e \in E,$$

composed by pullback. When C has finite products, $\mathsf{Span}(C, E)$ inherits a symmetric monoidal structure. In this language, a **three-functor formalism** on (C, E) is simply a lax symmetric monoidal functor

$$\mathsf{D} \colon \mathsf{Span}(C, E) \longrightarrow \mathsf{Cat},$$

which encodes f^* (backward legs), $f_!$ (forward legs), and $\otimes$ (monoidal structure) in a single coherent package. The remaining three operations f_*, $f^!$, $\underline{\mathrm{Hom}}$ are then obtained as right adjoints, and are therefore *properties* of D rather than additional structure.

The challenge is that constructing a lax monoidal functor out of $\mathsf{Span}(C, E)$ directly is hard. In practice, one starts from a much simpler piece of data, namely a functor

$$\mathsf{D}_0 \colon C^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathsf{Cat})$$

encoding only f^* and $\otimes$, and asks when it extends to $\mathsf{Span}(\mathcal{C}, E)$. The key input is a **suitable decomposition** of E in the sense of Mann [Man22, Def. A.5.9]: a pair of wide subcategories $I, P \subset E$ (thought of as "open/étale-like" and "proper-like" morphisms respectively) such that every $e \in E$ factors as $e = p \circ i$ with $i \in I$ and $p \in P$, subject to mild cancellation and truncation axioms. The extension theorem then reads: if $i^* = \mathsf{D}_0(i)$ admits a left adjoint $i_\sharp$ and $p^* = \mathsf{D}_0(p)$ admits a right adjoint p_* satisfying the expected base change and projection formulae—crucially including the mixed Beck–Chevalley equivalence $i_\sharp p'_* \simeq p_* i'_\sharp$—then D_0 admits an essentially unique extension to a three-functor formalism D, in which

$$f_! = p_* \, i_\sharp \qquad \text{for any factorization } f = p \circ i.$$

Recently, Cnossen–Lenz–Linskens [CLL25] gave a conceptually streamlined account of this extension theorem: they construct an 2-categorical enhancement $\mathsf{Span}_2(\mathcal{C}, E)_{P,I}$ of $\mathsf{Span}(\mathcal{C}, E)$—whose 2-morphisms are spans of spans with legs in P and I—and show that the inclusion $\mathcal{C}^{\mathrm{op}} \hookrightarrow \mathsf{Span}_2(\mathcal{C}, E)_{P,I}$ is the universal (I, P)-biadjointable functor. Mann's result becomes a formal corollary, and as a byproduct one obtains 2-categorical coherence data for D (in particular the adjoints $i_\sharp \dashv i^* \dashv i_*$ and $p^* \dashv p_*$ become part of the structure rather than auxiliary choices).

This is the framework in which we will place SH. Taking $\mathcal{C} = \mathsf{Sch}^{\mathrm{qcqs}}$, E = morphisms locally of finite presentation, I = étale morphisms, and P = proper morphisms, the six-functor formalism on SH will be obtained by verifying that the pullback functor $f^* \colon \mathsf{SH}(X) \to \mathsf{SH}(Y)$ and the smash product on SH satisfy precisely the hypotheses above. The deep input—smooth base change and proper base change for SH—is classical work of Morel–Voevodsky [MV99] and Ayoub [Ayo07] / Cisinski–Déglise [CD19]; the extension to the full six-functor package is then formal.

2.2. Construction of Motivic Spectra

This section gives a careful construction of the stable motivic category $\mathsf{SH}(S)$ as a universal $\otimes$-inversion, together with the accompanying technology—symmetric sequences, free algebras, Bousfield localization—needed both for this construction and for the construction of specific motivic spectra in subsequent sections. The treatment follows Robalo [Rob15], Hoyois [Hoy17], and Bachmann–Hoyois [BH21]; we have also drawn on the cleaner exposition of Annala–Iwasa [AI25] where appropriate.

Before entering the construction, let us briefly recall the setting established in Chapter 1. For a quasi-compact quasi-separated scheme S, we built $\mathsf{H}(S)$ as the full subcategory

$$\mathsf{H}(S) \;=\; \mathsf{Shv}_{\mathrm{Nis}}(\mathsf{Sm}_S) \cap \mathsf{PShv}_{\mathbb{A}^1}(\mathsf{Sm}_S) \;\subset\; \mathsf{PShv}(\mathsf{Sm}_S)$$

of presheaves (valued in anima) satisfying both Nisnevich descent and $\mathbb{A}^1$-invariance. The corresponding localization functor was denoted

$$\mathrm{L}_{\mathrm{mot}} \colon \mathsf{PShv}(\mathsf{Sm}_S) \longrightarrow \mathsf{H}(S),$$

and we showed (Remark 1.5.3) that it is computed as the transfinite alternation $\mathrm{L}_{\mathrm{mot}} \simeq (\mathrm{L}_{\mathrm{Nis}} \circ \mathrm{L}_{\mathbb{A}^1})^{\circ \mathbb{N}}$. The pointed variant $\mathsf{H}_*(S)$ was obtained by passing to pointed objects, and we defined the Tate object

$$\mathbb{T}_S \;:=\; \mathrm{cofib}\big(\infty \colon S \hookrightarrow \mathbb{P}^1_S\big) \;\in\; \mathsf{H}_*(S),$$

which—by Proposition 1.6.4—is equivalent to $S^1 \wedge \mathbb{G}_m = S^{2,1}$.

The goal of this section is to invert $\mathbb{T}_S$ under the smash product, producing a presentably symmetric monoidal category $\mathsf{SH}(S)$ that extends the sequence

$$\mathsf{PShv}(\mathsf{Sm}_S) \xrightarrow{\mathrm{L}_{\mathrm{mot}}} \mathsf{H}(S) \xrightarrow{(-)_+} \mathsf{H}_*(S) \longrightarrow \mathsf{SH}(S)$$

from Chapter 1 by one further universal construction.

2.2.1. The universal $\otimes$-inversion problem

Let $\mathcal{C} \in \mathrm{CAlg}(\mathsf{Pr}^L)$ be a presentably symmetric monoidal category and let $T \in \mathcal{C}$. We seek the initial presentably symmetric monoidal $\mathcal{C}$-algebra in which the image of T becomes $\otimes$-invertible: that is, the initial object of $\mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}$ whose structure functor $\mathcal{C} \to \mathcal{D}$ sends T to an invertible object.

Proposition 2.2.1 (Robalo [Rob15, Proposition 2.9]). *There is a smashing localization*

$$(-)[T^{-1}]\colon \mathrm{Mod}_{\mathcal{C}}(\mathsf{Pr}^L) \longrightarrow \mathrm{Mod}_{\mathcal{C}}(\mathsf{Pr}^L)$$

whose essential image is spanned by those $\mathcal{D}$ on which T acts as an equivalence. The localization $\mathcal{C}[T^{-1}]$ acquires a unique presentably symmetric monoidal structure for which $\mathcal{C} \to \mathcal{C}[T^{-1}]$ is symmetric monoidal, and $\mathcal{D}[T^{-1}] \simeq \mathcal{C}[T^{-1}] \otimes_{\mathcal{C}} \mathcal{D}$ for every $\mathcal{D} \in \mathrm{Mod}_{\mathcal{C}}(\mathsf{Pr}^L)$.

This existence result is abstract; it gives no workable model of $\mathcal{C}[T^{-1}]$. To obtain one, we pass through symmetric sequences. The key point is that a sequential colimit model—the telescope

$$\mathsf{Tel}_T(\mathcal{D}) \coloneqq \mathrm{colim}\big(\mathcal{D} \xrightarrow{T\otimes -} \mathcal{D} \xrightarrow{T\otimes -} \cdots\big)$$

—fails to give the correct answer in $\mathrm{CAlg}(\mathsf{Pr}^L)$ in general: the telescope encodes only the cyclic $\mathbb{Z}$-action on $T^{\otimes n}$, whereas the universal property demands the full symmetric-group action. We must therefore use a construction that remembers the Σ_n-actions.

2.2.2. Symmetric sequences and free presentable algebras

Definition 2.2.2 (Symmetric sequences). Let $\mathbb{F}$ denote the groupoid of finite sets and bijections. For $\mathcal{C} \in \mathrm{CAlg}(\mathsf{Pr}^L)$, the category of **symmetric sequences** in $\mathcal{C}$ is

$$\mathsf{SSeq}(\mathcal{C}) \coloneqq \mathsf{Fun}(\mathbb{F}, \mathcal{C}),$$

equipped with the Day convolution symmetric monoidal structure induced by the disjoint-union monoidal structure on $\mathbb{F}$.

A symmetric sequence may be pictured as a list $X = (X_0, X_1, X_2, \ldots)$ with each X_n carrying a Σ_n-action. The Day convolution product is

$$(X \otimes Y)_n \;=\; \bigoplus_{p+q=n} \mathrm{Ind}_{\Sigma_p \times \Sigma_q}^{\Sigma_n}\big(X_p \otimes Y_q\big),$$

and the unit is $\mathbf{1}_{\mathsf{SSeq}(\mathcal{C})} = (\mathbf{1}_{\mathcal{C}}, \varnothing, \varnothing, \ldots)$.

Proposition 2.2.3 (Free $\mathcal{C}$-algebra on one object). *The symmetric sequence*

$$G_{\mathcal{C}} \;\coloneqq\; (\varnothing, \mathbf{1}_{\mathcal{C}}, \varnothing, \varnothing, \ldots) \;\in\; \mathsf{SSeq}(\mathcal{C}),$$

concentrated in degree 1, exhibits $\mathsf{SSeq}(\mathcal{C})$ as the free presentably symmetric monoidal $\mathcal{C}$-algebra on one generator. Concretely, for every $\mathcal{D} \in \mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}$, evaluation at $G_{\mathcal{C}}$ induces an equivalence of anima

$$\mathrm{Hom}_{\mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}}\big(\mathsf{SSeq}(\mathcal{C}), \mathcal{D}\big) \xrightarrow{\sim} \iota\mathcal{D}.$$

Proof sketch. This is a presentable categorical version of the classical fact that SSeq is the free symmetric monoidal cocompletion of the discrete category with one object. The key input is that $\mathbb{F}$ is the free symmetric monoidal anima on one generator, and that Day convolution with values in a presentable $\mathcal{C}$-algebra preserves colimits componentwise. See [Rob15, Lemma 2.3.1] or [BH21, Proposition C.1] for a detailed proof. □

The next piece of technology allows us to generate more intricate algebras by specifying a relation.

Definition 2.2.4 (Free symmetric algebra). For $X \in \mathcal{C}$, write

$$\mathrm{Sym}(X) \;\coloneqq\; \bigoplus_{n\geq 0} \big(X^{\otimes n}\big)_{h\Sigma_n} \;\in\; \mathrm{CAlg}(\mathcal{C})$$

for the free commutative algebra generated by X. It satisfies the universal property

$$\mathrm{Hom}_{\mathrm{CAlg}(\mathcal{C})}(\mathrm{Sym}(X), A) \;\simeq\; \mathrm{Hom}_{\mathcal{C}}(X, A) \qquad \text{for every } A \in \mathrm{CAlg}(\mathcal{C}).$$

Proposition 2.2.5 (Module categories as free pointed algebras). *For $X \in \mathcal{C}$, the module category $\mathrm{Mod}_{\mathrm{Sym}(X)}(\mathcal{C}) \in \mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}$ is the initial presentably symmetric monoidal $\mathcal{C}$-algebra equipped with a morphism $X \to \mathbf{1}$ from (the image of) X to the structure unit. Concretely, for every $\mathcal{D} \in \mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}$,*

$$\mathrm{Hom}_{\mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}}\big(\mathrm{Mod}_{\mathrm{Sym}(X)}(\mathcal{C}), \mathcal{D}\big) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{D}}\big(X_{\mathcal{D}}, \mathbf{1}_{\mathcal{D}}\big),$$

where $X_{\mathcal{D}}$ denotes the image of X under the structure functor $\mathcal{C} \to \mathcal{D}$.

Proof sketch. By [Lur17, Theorem 4.8.5.16], maps of $\mathcal{C}$-algebras $\mathrm{Mod}_A(\mathcal{C}) \to \mathcal{D}$ correspond to maps of commutative algebras $A \to \mathbf{1}_{\mathcal{D}}$ in $\mathcal{C}$ (where $\mathbf{1}_{\mathcal{D}}$ is viewed as a $\mathcal{C}$-algebra via the structure functor). Applied to $A = \mathrm{Sym}(X)$, this becomes $\mathrm{Hom}_{\mathcal{C}}(X, \mathbf{1}_{\mathcal{D}})$, which under the structure functor identifies with $\mathrm{Hom}_{\mathcal{D}}(X_{\mathcal{D}}, \mathbf{1}_{\mathcal{D}})$. □

***Remark* 2.2.6.** The two universal properties (Propositions 2.2.3 and 2.2.5) are the only categorical inputs we will need. Everything that follows is a combination of these two and Bousfield localization.

2.2.3. Lax T-spectra

For $T \in \mathcal{C}$, consider the symmetric sequence

$$T_{\mathrm{sym}} \;:=\; (\varnothing, T, \varnothing, \varnothing, \ldots) \;\in\; \mathsf{SSeq}(\mathcal{C}),$$

concentrated in degree 1. Since $\mathsf{SSeq}(\mathcal{C})$ is itself a presentably symmetric monoidal $\mathcal{C}$-algebra, we may form its free commutative algebra $\mathrm{Sym}(T_{\mathrm{sym}}) \in \mathrm{CAlg}(\mathsf{SSeq}(\mathcal{C}))$, explicitly

$$\mathrm{Sym}(T_{\mathrm{sym}}) \;=\; \big(\mathbf{1}_{\mathcal{C}},\; T,\; (T^{\otimes 2})_{h\Sigma_2},\; (T^{\otimes 3})_{h\Sigma_3},\; \ldots\big),$$

with Σ_n-action on the n-th term given by the canonical permutation action.

Definition 2.2.7 (Lax T-spectra). The category of **lax T-spectra** in $\mathcal{C}$ is

$$\mathsf{Sp}_T^{\mathrm{lex}}(\mathcal{C}) \;:=\; \mathrm{Mod}_{\mathrm{Sym}(T_{\mathrm{sym}})}\big(\mathsf{SSeq}(\mathcal{C})\big) \;\in\; \mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}.$$

Combining Propositions 2.2.3 and 2.2.5, we obtain:

Proposition 2.2.8 (Universal property of $\mathsf{Sp}_T^{\mathrm{lex}}$). *For every $\mathcal{D} \in \mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}$ with structure functor $\varphi\colon \mathcal{C} \to \mathcal{D}$, there is a natural equivalence*

$$\mathrm{Hom}_{\mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}}\big(\mathsf{Sp}_T^{\mathrm{lex}}(\mathcal{C}), \mathcal{D}\big) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{D}}\big(\varphi(T), \mathbf{1}_{\mathcal{D}}\big).$$

That is, $\mathsf{Sp}_T^{\mathrm{lex}}(\mathcal{C})$ is the initial presentably symmetric monoidal $\mathcal{C}$-algebra equipped with a morphism $T \to \mathbf{1}$.

Explicitly, an object of $\mathsf{Sp}_T^{\mathrm{lex}}(\mathcal{C})$ is a symmetric sequence $E = (E_0, E_1, E_2, \ldots)$ together with Σ_n-equivariant **bonding maps**

$$T \otimes E_n \longrightarrow E_{n+1}$$

(equivalently, by $\otimes$-adjunction, maps $E_n \to \mathrm{Hom}(T, E_{n+1})$) satisfying the associativity and $\Sigma_\bullet$-coherence expected of modules over $\mathrm{Sym}(T_{\mathrm{sym}})$.

2.2.4. T-spectra as a Bousfield localization

The universal property of $\mathsf{Sp}_T^{\mathrm{lex}}$ specifies a morphism $T \to \mathbf{1}$ (equivalently, a structure bonding map), but does not force it to be an equivalence. To turn T into an invertible object we must perform a further localization.

Definition 2.2.9 (T-spectra). The category of **T-spectra** in $\mathcal{C}$ is the $\otimes$-Bousfield localization

$$\mathrm{L}_{\mathsf{Sp}}\colon \mathsf{Sp}_T^{\mathrm{lex}}(\mathcal{C}) \longrightarrow \mathsf{Sp}_T(\mathcal{C})$$

at the class of bonding maps $T \otimes E \to \sigma_+ E$, where σ_+ is the shift operator $(\sigma_+ E)_n = E_{n-1}$. A lax T-spectrum E is a *T-spectrum* if and only if the adjoint bonding maps

$$E_n \xrightarrow{\sim} \mathrm{Hom}(T, E_{n+1})$$

are equivalences for all n.

Theorem 2.2.10 (Robalo [Rob15, Corollary 2.22]). *$\mathsf{Sp}_T(\mathcal{C})$ realizes the formal inversion of T:*

$$\mathsf{Sp}_T(\mathcal{C}) \simeq \mathcal{C}[T^{-1}] \qquad \text{in } \mathrm{CAlg}(\mathsf{Pr}^L)_{\mathcal{C}/}.$$

In particular, it is initial among presentably symmetric monoidal $\mathcal{C}$-algebras in which T becomes $\otimes$-invertible.

Theorem 2.2.11 (Telescope comparison [Rob15, Corollary 2.22]). *If T is **symmetric** in the sense that the cyclic permutation $\sigma\colon T^{\otimes 3} \to T^{\otimes 3}$, $a \otimes b \otimes c \mapsto b \otimes c \otimes a$, is homotopic to the identity in $\mathcal{C}$, then the canonical comparison functor*

$$\mathsf{Sp}_T(\mathcal{C}) \xrightarrow{\sim} \mathsf{Tel}_T(\mathcal{C}) = \mathrm{colim}\big(\mathcal{C} \xrightarrow{T\otimes -} \mathcal{C} \xrightarrow{T\otimes -} \cdots\big)$$

is an equivalence.

2.2.5. Two adjunctions and the explicit description

We now unwind what $\mathsf{Sp}_T(\mathcal{C})$ looks like as a concrete category. The structure functor factors as

$$\mathcal{C} \xrightarrow{\Sigma^\infty_T} \mathsf{Sp}_T(\mathcal{C}) \xrightarrow{\Omega^\infty_T} \mathcal{C},$$

with $\Sigma^\infty_T \dashv \Omega^\infty_T$, where $\Omega^\infty_T E = E_0$ and $\Sigma^\infty_T X$ is the "suspension spectrum" with $(\Sigma^\infty_T X)_n = T^{\otimes n} \otimes X$. More generally, for each $n \geq 0$ there is an adjunction

$$\Sigma^{\infty-n}_T \dashv \Omega^{\infty-n}_T, \qquad \Omega^{\infty-n}_T E = E_n.$$

Since T is invertible in $\mathsf{Sp}_T(\mathcal{C})$, these adjunctions extend to all $n \in \mathbb{Z}$, and we may define $E(n) := \Omega^{\infty-n}_T E$ for every integer n. The structure of a T-spectrum is then captured by the following:

Proposition 2.2.12 (Explicit description of T-spectra). *The data of an object $E \in \mathsf{Sp}_T(\mathcal{C})$ is equivalent to:*

- *a sequence of objects $\big(E(n)\big)_{n\in\mathbb{Z}}$ in $\mathcal{C}$;*
- *equivalences $c_n\colon E(n) \xrightarrow{\sim} \mathrm{Hom}(T, E(n+1))$ for every $n \in \mathbb{Z}$.*

A commutative algebra structure on E corresponds further to the data of compatible graded multiplications $E(m) \otimes E(n) \to E(m+n)$; equivalently, $(E, c) \in \mathrm{CAlg}(\mathsf{Sp}_T(\mathcal{C}))$ with $\Omega^{\infty-n}_T(E) = E(n)$ for all $n \in \mathbb{Z}$.

This description will be the principal working definition of a motivic spectrum in the applications below.

2.2.6. From graded algebras to motivic spectra

A common source of commutative algebras in $\mathsf{Sp}_T(\mathcal{C})$—including the motivic K-theory spectrum KGL and its relatives—is graded commutative algebras in $\mathcal{C}$ equipped with a distinguished *Bott element*. This subsection formalizes that passage.

Construction 2.2.13 (Graded algebras via $\mathsf{Fun}(\mathbb{Z}, \mathcal{C})$). Equip the set $\mathbb{Z}$ with its discrete groupoid structure, so that $\mathsf{Fun}(\mathbb{Z}, \mathcal{C}) \simeq \prod_{n\in\mathbb{Z}} \mathcal{C}$ acquires a symmetric monoidal structure via Day convolution for the addition operation on $\mathbb{Z}$. A **$\mathbb{Z}$-graded commutative algebra** in $\mathcal{C}$ is an element of $\mathrm{CAlg}(\mathsf{Fun}(\mathbb{Z}, \mathcal{C}))$; concretely, a sequence $E(*) = (E(n))_{n\in\mathbb{Z}}$ with graded multiplications $E(m) \otimes E(n) \to E(m+n)$ satisfying commutativity and associativity coherently.

The inclusion of groupoids $\mathbb{F} \hookrightarrow \mathbb{N} \hookrightarrow \mathbb{Z}$ (with $\mathbb{F}$ the groupoid of finite sets, $\mathbb{N}$ the discrete monoid of natural numbers) induces a lax symmetric monoidal functor

$$\mathsf{Fun}(\mathbb{Z}, \mathcal{C}) \xrightarrow{\text{forget}} \mathsf{SSeq}(\mathcal{C}),$$

restricting a graded object to its underlying symmetric sequence (forgetting the Σ_n-actions, which are trivial in the $\mathbb{Z}$-graded case).

Proposition 2.2.14 (Graded algebras with Bott element give lax spectra). *Let $E(*) \in \mathrm{CAlg}(\mathsf{Fun}(\mathbb{Z}, \mathcal{C}))$ be a $\mathbb{Z}$-graded commutative algebra in $\mathcal{C}$, and let $c\colon T \to E(1)$ be a morphism in $\mathcal{C}$ (a **Bott element**). Then the pair $(E(*), c)$ canonically promotes to a commutative algebra*

$$(E, c) \in \mathrm{CAlg}\big(\mathsf{Sp}_T^{\mathrm{lex}}(\mathcal{C})\big),$$

with underlying symmetric sequence the restriction of $E(*)$ and bonding maps $T \otimes E(n) \xrightarrow{c \otimes 1} E(1) \otimes E(n) \to E(n+1)$ given by multiplication.

Proof sketch. The Bott element c and the graded multiplication together determine a morphism $T_{\mathrm{sym}} \to E$ in $\mathsf{SSeq}(\mathcal{C})$ (using the forgetful functor from $\mathsf{Fun}(\mathbb{Z}, \mathcal{C})$); the universal property of $\mathrm{Sym}(T_{\mathrm{sym}})$ then makes E into a $\mathrm{Sym}(T_{\mathrm{sym}})$-module, i.e., a lax T-spectrum. □

The spectrum (E, c) constructed above lies in $\mathsf{Sp}_T^{\mathrm{lex}}$; to promote it to an honest T-spectrum, one needs the bonding maps to become equivalences. This is not automatic—it is a property of $(E(*), c)$—and one typically enforces it via the Bousfield localization of the next subsection.

2.2.7. Bott-periodic algebras

We now describe an alternative construction, due in this form to Bachmann–Hoyois [BH21], which directly produces objects of $\mathsf{Sp}_T(\mathcal{C})$ from pairs (E, β) where β is a Bott element. This construction is especially convenient when E is defined as a presheaf on schemes and one wishes to promote it to a motivic spectrum.

Definition 2.2.15 (Bott-periodic modules). Let $E \in \mathrm{CAlg}(\mathcal{C})$ and let $\beta\colon T \to E$ be a morphism in $\mathcal{C}$ (the **Bott element**). The adjoint of β is a morphism $\mathbf{1} \to \mathrm{Hom}(T, E)$, equivalently a morphism $E \to \mathrm{Hom}(T, E)$ of E-modules. The category of **β-periodic E-modules** is the $\otimes$-Bousfield localization

$$\mathcal{P}_\beta \mathrm{Mod}_E(\mathcal{C}) \subset \mathrm{Mod}_E(\mathcal{C})$$

spanned by those modules M for which the structure map $M \to \mathrm{Hom}(T, M)$ is an equivalence (equivalently, those on which $T \otimes -$ acts as an equivalence).

Proposition 2.2.16 (Promotion to T-spectra [BH21, §4]). *Let $E \in \mathrm{CAlg}(\mathcal{C})$ and $\beta\colon T \to E$ be as above. Then there is a natural equivalence*

$$\mathcal{P}_\beta \mathrm{Mod}_E(\mathcal{C}) \simeq \mathcal{P}_\beta \mathrm{Mod}_E\big(\mathsf{Sp}_T(\mathcal{C})\big),$$

where on the right E is viewed as a commutative algebra in $\mathsf{Sp}_T(\mathcal{C})$ via its image under Σ_T^∞. Consequently, E admits an essentially unique promotion to a commutative algebra

$$E \in \mathrm{CAlg}\big(\mathsf{Sp}_T(\mathcal{C})\big) \qquad \text{with } \Omega_T^{\infty - n} E = E \text{ for all } n \in \mathbb{Z}.$$

Informally: the Bott element β already encodes enough "symmetry" to force E to become T-periodic, and this periodicity is exactly what is needed for E to lift from $\mathcal{C}$ to $\mathsf{Sp}_T(\mathcal{C})$.

2.2.8. Specialization to motivic homotopy theory

We now apply the preceding machinery to the setting of motivic homotopy theory. The input data is the pointed motivic homotopy category $\mathsf{H}_*(S)$ from Chapter 1, equipped with the smash product, and the Tate object $\mathbb{T}_S \in \mathsf{H}_*(S)$. To use the machinery of this section, the only remaining verification is that $\mathbb{T}_S$ is symmetric in the sense of Theorem 2.2.11.

Proposition 2.2.17 ($\mathbb{T}_S$ is symmetric). *The cyclic permutation $\sigma\colon \mathbb{T}_S^{\wedge 3} \to \mathbb{T}_S^{\wedge 3}$ is $\mathbb{A}^1$-homotopic to the identity in $\mathsf{H}_*(S)$.*

Proof. By Proposition 1.6.4, $\mathbb{T}_S \simeq \mathbb{P}^1_S$. It suffices to show that a transposition on $\mathbb{P}^1 \wedge \mathbb{P}^1$ is $\mathbb{A}^1$-homotopic to $-\mathrm{id}$, for then a 3-cycle in Σ_3—being a composite of two transpositions—acts as $(-\mathrm{id})^2 = \mathrm{id}$.

Consider the action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathbb{P}^1 \wedge \mathbb{P}^1$ via its action on the two factors. We claim that the two matrices

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad \text{and} \qquad \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

are $\mathbb{A}^1$-homotopic. Indeed, set

$$A \coloneqq \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix},$$

and since $A \in \mathrm{SL}_2(\mathbb{Z})$ factors as a product of elementary matrices (each of which is $\mathbb{A}^1$-homotopic to the identity via $\begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$, $t \in \mathbb{A}^1$), A itself is $\mathbb{A}^1$-homotopic to the identity. Thus the transposition τ on $\mathbb{P}^1 \wedge \mathbb{P}^1$ is $\mathbb{A}^1$-homotopic to $(-\mathrm{id}) \wedge \mathrm{id} \simeq -\mathrm{id}$, as claimed. □

Definition 2.2.18 (Stable motivic category). The **stable motivic category** over S is

$$\mathsf{SH}(S) \coloneqq \mathsf{Sp}_{\mathbb{T}_S}\big(\mathsf{H}_*(S)\big) = \mathsf{H}_*(S)\big[\mathbb{T}_S^{-1}\big] \in \mathrm{CAlg}(\mathsf{Pr}^L_{\mathrm{st}}).$$

By Theorem 2.2.11 and Proposition 2.2.17, it admits the explicit telescope model

$$\mathsf{SH}(S) \simeq \operatorname{colim}\Big(\mathsf{H}_*(S) \xrightarrow{-\wedge \mathbb{T}_S} \mathsf{H}_*(S) \xrightarrow{-\wedge \mathbb{T}_S} \cdots\Big).$$

Assigning $S \mapsto \mathsf{SH}(S)$ and $f : T \to S$ to the symmetric monoidal pullback $f^* \colon \mathsf{SH}(S) \to \mathsf{SH}(T)$ defines a functor

$$\mathsf{SH} \colon (\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathsf{Pr}^L_{\mathrm{st}}).$$

***Remark* 2.2.19** (Comparison with Chapter 1's construction chain). Putting together all the constructions from Chapter 1 and this section, we obtain the full sequence of symmetric monoidal localizations

$$\mathsf{PShv}(\mathsf{Sm}_S) \xrightarrow{\mathrm{L}_{\mathrm{mot}}} \mathsf{H}(S) \xrightarrow{(-)_+} \mathsf{H}_*(S) \xrightarrow{\Sigma^\infty_{\mathbb{T}}} \mathsf{SH}(S),$$

each realizing a universal construction in $\mathrm{CAlg}(\mathsf{Pr}^L)$:

- $\mathrm{L}_{\mathrm{mot}}$: free enforcement of Nisnevich descent and $\mathbb{A}^1$-invariance;
- $(-)_+$: free adjunction of a basepoint;
- $\Sigma^\infty_{\mathbb{T}}$: free inversion of $\mathbb{T}_S$ under the smash product.

Each stage is determined by a clean universal property, and the composition is the universal symmetric monoidal stable category attached to Sm_S that is Nisnevich-local, $\mathbb{A}^1$-invariant, and $\mathbb{T}$-stable. This decomposition is crucial for the construction of norm functors (see the end of Chapter 1): each of the three arrows is a universal construction in $\mathrm{CAlg}(\mathsf{Cat}^{\mathrm{sift}})$, so a functor defined at the level of $\mathsf{Sm}_{S,+}$ has at most one sifted-colimit-preserving symmetric monoidal extension at each stage.

***Remark* 2.2.20** (Stability). Since $\mathbb{T}_S \simeq S^1 \wedge \mathbb{G}_m$ in $\mathsf{H}_*(S)$, inverting $\mathbb{T}_S$ in particular inverts the simplicial circle S^1 (after one smash with $\mathbb{G}_m$); hence $\mathsf{SH}(S)$ is stable. The bigraded spheres $S^{p,q} = (S^1)^{\wedge(p-q)} \wedge \mathbb{G}_m^{\wedge q}$ from §1.6.2 refine the usual integer suspension, and both $\Sigma^{1,0}$ (simplicial) and $\Sigma^{1,1}$ (Tate) are invertible.

2.2.9. Example: the motivic K-theory spectrum KGL

We illustrate the Bott-periodic construction of §2.2.7 by building the motivic K-theory spectrum $\mathrm{KGL}_S \in \mathrm{CAlg}(\mathsf{SH}(S))$. This serves both as the prototype for the method and as a foundational object we will return to repeatedly.

Let $\mathrm{K} \colon \mathsf{Sm}_S^{\mathrm{op}} \to \mathsf{Sp}$ denote non-connective algebraic K-theory. Since K fails $\mathbb{A}^1$-invariance on singular schemes (Remark 1.5.8), we first replace it by homotopy K-theory

$$\mathrm{KH}(X) \coloneqq \mathrm{L}_{\mathbb{A}^1}\mathrm{K}(X) = \operatorname{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathrm{K}\big(X \times \Delta^n\big),$$

a Nisnevich sheaf of spectra on $\mathsf{Sch}^{\mathrm{qcqs}}$ satisfying $\mathbb{A}^1$-invariance. Restricted to Sm_S, KH defines an object of $\mathrm{CAlg}(\mathsf{H}(S); \mathsf{Sp})$ (the spectrum-valued analogue of $\mathsf{H}(S)$; we omit the notational refinement).

The Bott element arises geometrically: the fiber sequence $\mathcal{O}(-1) \to \mathcal{O} \to \mathcal{O}_\infty$ of vector bundles on $\mathbb{P}^1$ determines a class

$$\beta \in \mathrm{fib}\big(\mathrm{K}(\mathbb{P}^1) \to \mathrm{K}(\infty)\big),$$

the class of $[\mathcal{O}(-1)] - [\mathcal{O}]$. Using $\mathbb{T}_S \simeq \mathbb{P}^1_S/\{\infty\}$, this yields a morphism

$$\beta \colon \mathbb{T}_S \longrightarrow \mathrm{KH}$$

in $\mathsf{H}_*(S)$ (after passing to spectrum-valued presheaves).

Theorem 2.2.21 ([BH21, §16]). *The pair* (KH, β) *is Bott-periodic, and its promotion* $\mathrm{KGL}_S \in \mathrm{CAlg}(\mathsf{SH}(S))$ *via Proposition 2.2.16 is the* ***motivic K-theory spectrum****. It satisfies* $\Omega^{\infty-n}_{\mathbb{T}}\mathrm{KGL}_S \simeq \mathrm{KH}$ *for all* $n \in \mathbb{Z}$*, reflecting the Bott periodicity* $\Sigma_{\mathbb{T}}\mathrm{KGL}_S \simeq \mathrm{KGL}_S$.

Proposition 2.2.22 (Base change for KGL). *For any morphism of schemes* $f : Y \to X$*, the natural map*

$$f^*\mathrm{KGL}_X \xrightarrow{\sim} \mathrm{KGL}_Y$$

is an equivalence in $\mathsf{SH}(Y)$.

Proof sketch. TBA. □

2.3. The Six-Functor Formalism on SH

Having constructed SH as a functor $(\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \to \mathrm{CAlg}(\mathsf{Pr}^L_{\mathrm{st}})$, we now extend it to a full six-functor formalism. We begin by recalling the relevant axioms.

2.3.1. Localization and base change

Definition 2.3.1 (Localization axiom). A functor $\mathsf{D} \colon (\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \to \mathsf{Cat}_{\mathrm{st}}$ satisfies the **localization axiom** if for every $X \in \mathsf{Sch}^{\mathrm{qcqs}}$, every closed immersion $i \colon Z \hookrightarrow X$ with open complement $j \colon U \hookrightarrow X$, the diagram

$$\begin{tikzcd} \mathsf{D}(Z) \arrow[r, "i_*"] & \mathsf{D}(X) \arrow[l, bend right, "i^*"'] \arrow[l, bend left, "i^!"] \arrow[r, "j^*"] & \mathsf{D}(U) \arrow[l, bend right, "j_!"'] \arrow[l, bend left, "j_*"] \end{tikzcd}$$

is a recollement. That is, (i^*, j^*) is conservative, and both i_* and j_* are fully faithful.

Definition 2.3.2 (Base change properties). Let $\mathsf{D} \colon (\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \to \mathsf{Cat}_{\mathrm{st}}$ be a functor.

- We say a morphism $g \colon C \to D$ satisfies **left base change** if for every pullback square

$$\begin{tikzcd} A \arrow[r, "k"] \arrow[d, "h"'] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, "f"] \\ C \arrow[r, "g"'] & D \end{tikzcd}$$

  the functor $k^* \colon \mathsf{D}(B) \to \mathsf{D}(A)$ admits a left adjoint $k_\sharp$, and the canonical morphism $k_\sharp h^* \to f^* g_\sharp$ is an isomorphism.

- We say a morphism $f \colon B \to D$ satisfies **right base change** if for every pullback square as above, the functor $h^* \colon \mathsf{D}(C) \to \mathsf{D}(A)$ admits a right adjoint h_*, and the canonical morphism $g^* f_* \to h_* k^*$ is an isomorphism.

Proposition 2.3.3. *Let* $\mathsf{D} \colon (\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \to \mathsf{Cat}_{\mathrm{st}}$ *satisfy the localization axiom.*

(i) If étale morphisms satisfy left base change, then D *satisfies Nisnevich descent.*

(ii) If finitely presented proper morphisms satisfy right base change, then D *satisfies abstract blowup descent.*

Proof. We treat the case of abstract blowup descent; the Nisnevich case is similar. Consider an abstract blowup square

$$\begin{tikzcd} E \arrow[r, "k"] \arrow[d, "q"'] \arrow[dr, phantom, "\lrcorner", very near start] & \tilde{X} \arrow[d, "p"] \\ Z \arrow[r, hook, "i"'] & X \end{tikzcd}$$

We must show that

$$\mathsf{D}(X) \xrightarrow{(p^*, i^*)} \mathsf{D}(\tilde{X}) \times_{\mathsf{D}(E)} \mathsf{D}(Z)$$

is an equivalence. The functor (p^*, i^*) admits a right adjoint

$$R\colon \mathsf{D}(\tilde{X}) \times_{\mathsf{D}(E)} \mathsf{D}(Z) \to \mathsf{D}(X), \quad (A \xrightarrow{\alpha} C \xleftarrow{\beta} B) \mapsto p_*A \times_{(iq)_*C} i_*B.$$

To show $(p^*, i^*) \dashv R$ is an adjoint equivalence, it suffices to verify that (p^*, i^*) is conservative and that the counit is an isomorphism.

Conservativity. Let $j\colon U \hookrightarrow X$ be the open complement of i. By the localization axiom, (i^*, j^*) is conservative. Since p restricts to an isomorphism over U, we have $j^* \simeq h^*p^*$ where $h\colon U \hookrightarrow \tilde{X}$ is the inclusion. Thus (p^*, i^*) is conservative.

Counit is an isomorphism. Let $(A \xrightarrow{\alpha} C \xleftarrow{\beta} B) \in \mathsf{D}(\tilde{X}) \times_{\mathsf{D}(E)} \mathsf{D}(Z)$, where $\alpha\colon k^*A \xrightarrow{\sim} C$ and $\beta\colon q^*B \xrightarrow{\sim} C$ are equivalences in $\mathsf{D}(E)$. We verify the counit componentwise.

For the i^*-component: by right base change for p along i, and the identification $iq = pk$,

$$i^*p_*A \simeq q_*k^*A \simeq q_*C,$$

using α for the second equivalence. Combined with $i^*i_*B \simeq B$ (full faithfulness of i_* from the localization axiom) and $i^*(iq)_*C \simeq q_*C$, the fiber product becomes

$$i^*\big(p_*A \times_{(iq)_*C} i_*B\big) \simeq q_*C \times_{q_*C} B \simeq B.$$

For the p^*-component: we must show $p^*(p_*A \times_{(iq)_*C} i_*B) \xrightarrow{\sim} A$. Let $h\colon U \hookrightarrow \tilde{X}$ be the pullback of j along p. Since the abstract blowup square is a pullback, h is also the open complement of k. By the localization axiom on $\tilde{X}$, (k^*, h^*) is conservative, so it suffices to verify the equivalence after applying k^* and h^* separately.

Applying k^*: by right base change,

$$k^*p^*(p_*A \times_{(iq)_*C} i_*B) \simeq q^*i^*(p_*A \times_{(iq)_*C} i_*B) \simeq q^*B \simeq C \simeq k^*A,$$

where the second equivalence uses the i^*-component computation above.

Applying h^*: since i_*B and $(iq)_*C$ are supported on Z, applying $j^* = h^*p^*$ kills them, yielding j^*p_*A. We must show $j^*p_*A \simeq h^*A$. Consider the commutative diagram of fiber sequences from the localization axiom:

$$\begin{tikzcd} i_*i^!p_*A \arrow[r] \arrow[d] & p_*A \arrow[r] \arrow[d, equal] & j_*j^*p_*A \arrow[d] \\ p_*k_*k^!A \arrow[r] & p_*A \arrow[r] & p_*h_*h^*A \end{tikzcd}$$

By right base change for i along p, the left vertical morphism is an equivalence, hence so is the right one. Since $p_*h_* = (ph)_* = j_*$ and j_* is fully faithful, this gives $j^*p_*A \simeq h^*A$. □

2.3.2. The six-functor formalism for SH

Theorem 2.3.4 (Morel–Voevodsky [MV99]). *The stable motivic functor* $\mathsf{SH}\colon (\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \to \mathsf{Pr}^L_{\mathrm{st}}$ *satisfies the localization axiom.*

Theorem 2.3.5 (Ayoub [Ayo07], Cisinski–Déglise [CD19]). *There is an essentially unique lax symmetric monoidal extension making the following diagram commute:*

$$\begin{tikzcd} (\mathsf{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \arrow[r, "\mathsf{SH}"] \arrow[d, hook] & \mathrm{CAlg}(\mathsf{Pr}^L_{\mathrm{st}}) \\ \mathsf{Span}_2(\mathsf{Sch}^{\mathrm{qcqs}}, E)^{\otimes}_{P,I} \arrow[ur, dashed] & \end{tikzcd}$$

with E = morphisms locally of finite presentation, I = étale morphisms, P = proper morphisms. The $(\infty, 2)$-category $\mathsf{Span}_2(\mathsf{Sch}^{\mathrm{qcqs}}, E)_{P,I}$ has:

- objects: *quasi-compact quasi-separated schemes;*
- 1-morphisms $X \to Z$: *spans*

$$\begin{tikzcd} & Y \arrow[dl, "f"'] \arrow[dr, "g"] & \\ X & & Z \end{tikzcd}$$

  with f of finite presentation and g locally of finite presentation;

- 2-morphisms *between two such spans: diagrams*

$$\begin{tikzcd} & Y & \\ X \arrow[ur] & Y'' \arrow[u, "a"] \arrow[d, "b"] & Z \arrow[ul] \\ & Y' \arrow[ul] \arrow[ur] & \end{tikzcd}$$

  with a proper and b étale.

*In this extension, the **exceptional pushforward** along $f \in E$ is given by $f_! = p_* i_\sharp$ for any factorization $f = p \circ i$ with i étale and p proper.*

***Remark* 2.3.6** (Structure of the proof). The proof is not a monolith; it decomposes into a formal step and a deep geometric step.

The *formal step* is the universal property of $\mathsf{Span}_2(-, E)^{\otimes}_{P,I}$ established by Cnossen–Lenz–Linskens [CLL25, Thm. 5.17]: given any lax symmetric monoidal functor $\mathsf{D}_0 \colon (\mathcal{C}^{\times})^{\mathrm{op}} \to \mathsf{Cat}^{\times}$ which is (I, P)-biadjointable and satisfies the projection formula, there is an essentially unique extension to $\mathsf{Span}_2(\mathcal{C}, E)^{\otimes}_{P,I}$. This step is purely 2-categorical and contains no algebro-geometric content.

The *geometric step* is to verify the input hypothesis, namely that SH is (I, P)-biadjointable. This breaks further into three ingredients:

- **Smooth/étale base change and the projection formula for $i_\sharp$:** these follow essentially from the construction of SH, since the smooth pushforward $i_\sharp$ on $\mathsf{PShv}(\mathsf{Sm}_-)$ is computed via the restriction along $\mathsf{Sm}_{/T} \to \mathsf{Sm}_{/S}$ for $i \colon T \to S$ étale, and this is compatible with base change at the level of presheaves.
- **Localization:** the recollement (i^*, j^*) for closed immersions i with open complement j. This is Theorem 2.3.4, due to Morel–Voevodsky [MV99].
- **Proper base change and the projection formula for p_*:** this is the deep input. Ayoub [Ayo07] proved proper base change for projective morphisms; combined with localization [MV99], this yields proper base change for all projective morphisms. Cisinski–Déglise [CD19] extended this to arbitrary proper morphisms using Chow's lemma and a Noetherian approximation argument.

Once all three ingredients are in hand, the full six-functor package on SH is obtained by invoking the universal property [CLL25]. The mixed Beck–Chevalley $i_\sharp p'_* \simeq p_* i'_\sharp$ is not an additional input: it follows from proper base change applied to a pullback square whose one edge is an étale morphism.

By Proposition 2.3.3 combined with Theorem 2.3.4 and the proper base change part of Remark 2.3.6, we obtain:

Corollary 2.3.7. *The functor* SH *is a cdh sheaf.*

Corollary 2.3.8 (Cisinski)**.** KH *is a cdh sheaf on* $\mathsf{Sch}^{\mathrm{qcqs}}$.

Proof sketch. Combine Proposition 2.2.22 with Corollary 2.3.7: the base-change compatibility of KGL shows that $\mathrm{KH}(X) = \mathrm{Hom}_{\mathsf{SH}(X)}(\mathbf{1}, \mathrm{KGL}_X)$, and cdh descent for SH then forces cdh descent on KH. □

Bibliography

[AHW17] Aravind Asok, Marc Hoyois, and Matthias Wendt. Affine representability results in $\mathbb{A}^1$-homotopy theory, I: vector bundles. *Duke Mathematical Journal*, 166(10):1923–1953, 2017.

[AI25] Toni Annala and Ryomei Iwasa. Motivic spectra and universality of k-theory, 2025.

[Ayo07] Joseph Ayoub. *Les six opérations de Grothendieck et le formalisme des cycles évanescents dans le monde motivique*. PhD thesis, Université Paris Diderot, 2007. Published as Astérisque **314–315** (2007).

[BEM25] Tom Bachmann, Elden Elmanto, and Matthew Morrow. $\mathbb{A}^1$-invariant motivic cohomology of schemes. 2025.

[BH21] Tom Bachmann and Marc Hoyois. Norms in motivic homotopy theory. *Astérisque*, 425:ix+207, 2021.

[CD19] Denis-Charles Cisinski and Frédéric Déglise. *Triangulated Categories of Mixed Motives*. Springer Monographs in Mathematics. Springer, 2019.

[CLL25] Bastiaan Cnossen, Tobias Lenz, and Sil Linskens. Universality of span 2-categories and the construction of 6-functor formalisms. 2025.

[DG22] Brad Drew and Martin Gallauer. The universal six-functor formalism. *Annals of K-Theory*, 7(4):599–649, December 2022.

[Hoy] Marc Hoyois. The homotopy category of an interval object. Unpublished notes.

[Hoy14] Marc Hoyois. A quadratic refinement of the Grothendieck–Lefschetz–Verdier trace formula. *Algebraic & Geometric Topology*, 14(6):3603–3658, 2014.

[Hoy17] Marc Hoyois. The six operations in equivariant motivic homotopy theory. *Advances in Mathematics*, 305:197–279, January 2017.

[Lur17] Jacob Lurie. *Higher Algebra*. 2017.

[LZ24] Yifeng Liu and Weizhe Zheng. Enhanced six operations and base change theorem for higher artin stacks, 2024.

[Man22] Lucas Mann. A p-adic 6-functor formalism in rigid-analytic geometry, 2022.

[Mor12] Fabien Morel. *$\mathbb{A}^1$-Algebraic Topology over a Field*, volume 2052 of *Lecture Notes in Mathematics*. Springer, 2012.

[MV99] Fabien Morel and Vladimir Voevodsky. $\mathbb{A}^1$-homotopy theory of schemes. *Publications Mathématiques de l'IHÉS*, 90:45–143, 1999.

[Qui76] Daniel Quillen. Projective modules over polynomial rings. *Inventiones Mathematicae*, 36:167–171, 1976.

[Rob15] Marco Robalo. K-theory and the bridge from motives to noncommutative motives. *Advances in Mathematics*, 269:399–550, 2015.

[Sch26] Peter Scholze. Six-functor formalisms, 2026.

[Ser55] Jean-Pierre Serre. Faisceaux algébriques cohérents. *Annals of Mathematics*, 61(2):197–278, 1955.

[Sus76] Andrei A. Suslin. Projective modules over polynomial rings are free. *Doklady Akademii Nauk SSSR*, 229(5):1063–1066, 1976.